MIL-M-28787/109B(EC)
4 February 1983
SUPERSEDING
MIL-M-28787/109A(EC)
9 November 1978

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, INTERFACE, DIGITAL, KEY CODE MDL

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface driver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 3.57 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MDL.

3.5.2 Size. The module size shall be 1A .

3.5.3 Weight. The module shall have a maximum weight of 85 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40 .

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/109-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

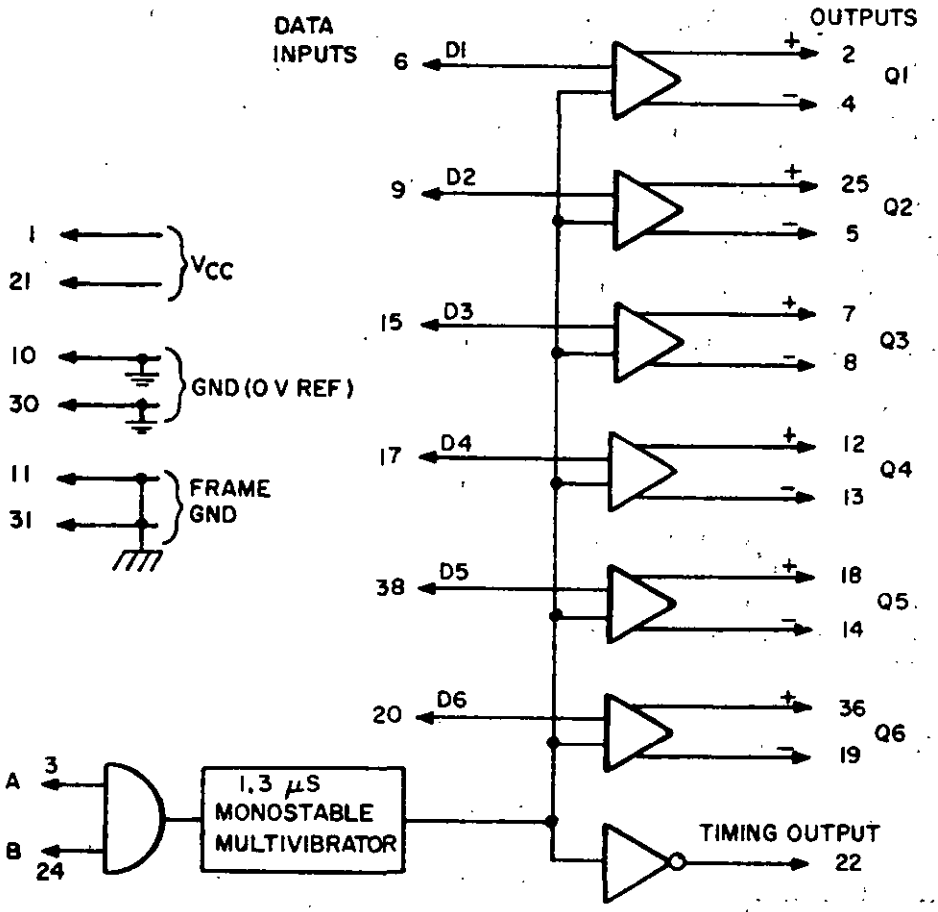
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Unused pins = 16, 23, 26-29, 32-35, 37, 39, and 40.

FIGURE 1. MDL function diagram, interface driver.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A3	Isolation	II	X	10	∞	10	∞	10	∞	10	∞	MΩ
B1-B7	Continuity	II	X	0	5	0	5	0	5	0	5	Ω
B8	Continuity	II	Q	0	10	0	10	0	10	0	10	Ω
C1-C6, C15-C20	Output noise voltage	III	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
C7,C8	Timing output pulse amplitude	III	X	3.8	5.5	3.8	5.5	3.8	5.5	3.8	5.5	V
C9-C14	Driver output pulse amplitude	III	X	3.2	5.0	3.2	5.5	3.2	5.0	3.2	5.5	V
---	Output pulse characteristics											
E1-E6	Pulse amplitude	IV	X	3.2	5.5	3.2	5.5	3.2	5.5	3.2	5.5	V
E1-E6	Pulse width	IV	X	1.10	1.45	1.10	1.45	1.10	1.45	1.10	1.45	µs
E1-E6	Transition time (t_{TLH})	IV	X	5	52	5	52	5	60	5	60	ns
E1-E6	Transition time (t_{THL})	IV	X	10	70	10	70	8	70	8	70	ns
E1-E6	Droop	IV	X	-0.3	0.3	-0.3	0.3	-0.3	0.3	-0.3	0.3	V
E1-E6	Undershoot	IV	X	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	V
E1-E6	Turn on time	IV	X	5	220	5	220	5	220	5	220	ns
---	Timing output pulse characteristics											
E7,E8	Pulse amplitude	IV	X	3.8	5.5	3.8	5.5	3.8	5.5	3.8	5.5	V
E7,E8	Pulse width	IV	X	1.10	1.45	1.10	1.45	1.10	1.45	1.10	1.45	µs
E7,E8	Transition time (t_{TLH})	IV	X	5	30	5	30	5	30	5	30	ns
E7,E8	Transition time (t_{THL})	IV	X	100	350	100	350	100	350	100	350	ns

---

See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
E7,E8	Turn on time	IV	X	5	220	5	220	5	220	5	220	ns
E9	V_{CE} (SAT)	IV	X	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	V
---	Output pulse characteristics											
F1	Pulse amplitude	IV	S	3.5	5.5	3.5	5.5	3.5	5.5	3.5	5.5	V
F1	Pulse width	IV	S	1.10	1.45	1.10	1.45	1.10	1.45	1.10	1.45	µs
F1	Transition time (t_{TLH})	IV	S	---	---	---	---	---	---	---	---	ns
F1	Transition time (t_{THL})	IV	S	---	---	---	---	---	---	---	---	ns
F1	Droop	IV	S	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
F1	Undershoot	IV	S	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	V
F1	Turn on time	IV	S	5	220	5	220	5	220	5	220	ns
---	Timing output pulse characteristics											
F2	Pulse amplitude	IV	S	3.8	5.5	3.8	5.5	3.8	5.5	3.8	5.5	V
F2	Pulse width	IV	S	1.10	1.45	1.10	1.45	1.10	1.45	1.10	1.45	µs
F2	Transition time (t_{TLH})	IV	S	5	30	5	30	5	30	5	30	ns
F2	Transition time (t_{THL})	IV	S	100	350	100	350	100	350	100	350	ns
F2	Droop	IV	S	---	---	---	---	---	---	---	---	V
F2	Undershoot	IV	S	---	---	---	---	---	---	---	---	V
F2	Turn on time	IV	S	5	220	5	220	5	220	5	220	ns
F3	V_{CE} (SAT)	IV	S	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	V

See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
G1,G2	Input current, timing, low level	V	X	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	mA
H1,H2	Input current, timing, high level	V	X	0.0	2.0	0.0	4.0	0.0	3.0	0.0	5.0	µA
J1-J6	Input current, data, high level	V	X	0.0	4.0	0.0	8.0	0.0	6.0	0.0	10.0	µA
K1-K6	Input voltage, data, low level	V	X	0.00	0.32	0.00	0.32	0.00	0.32	0.00	0.32	V
L	Output short circuit current	V	X	0.7	3.0	0.7	3.0	0.7	3.0	0.7	3.0	mA
M1	Power supply current	VI	S	0	50	0	50	0	50	0	50	mA
M2	Power supply current	VI	S	0	150	0	150	0	150	0	150	mA
N	Zener applied potential	VII	S	0	150	0	150	0	150	0	150	mA
P1-P8	Reverse voltage	VIII	S	3.8	5.5	3.8	5.5	3.8	5.5	3.8	5.5	V
Q	V_{OL} loaded output	IX	S	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	V
R1-R8	Input capacitance	X	S	0	20	0	20	0	20	0	20	pF
S1,S2	Input transient immunity	XI	S	2.0	5.5	2.0	5.5	2.0	5.5	2.0	5.5	V
S3	Input transient immunity	XI	S	0	300	0	300	0	300	0	300	ns
T1-T13	Input crosstalk	XII	Q	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
T14	Input Crosstalk	XII	Q	2	5	2	5	2	5	2	5	V
U	Power supply filter capacitance	XIII	S	0.10	0.26	0.10	0.26	0.10	0.26	0.10	0.26	µF
V	Output short and open	XIV	S	0	150	0	150	0	150	0	150	mA
W1-W3	Imbalance current	XV	S	0	40	0	40	0	40	0	40	mA

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25° C.
3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 20260[illegible]-20T16:03Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions M through W, F, A through E, or G through L are part of a sequence of tests, the order of testing shall be as follows: M through W, F, A through E, G through L where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through E and G through M shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

*6.3 Cross reference of classification. Military Specification MIL-M-28787/109 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658832. Part number M28787/109-1 supersedes part numbers 2658832 and M28787/109.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage	5.0 V dc ±10 percent
Maximum power dissipation	
Operating	825 mW
Quiescent	275 mW
Speed	DC to 167 kHz maximum

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this.is the current version before use.

Input voltage	
High level (V_{IH})	2.0 V minimum, V_{CC} maximum
Low level (V_{IL})	0 V minimum, 1.2 V maximum
Input current	
High Level (I_{IH})	10 µA maximum
Low Level (I_{IL})	-1.6 mA maximum
Timing output voltage	
High level	3.8 V minimum, V_{CC} maximum
Low level	0 V minimum, 0.45 V maximum
Output drive current	
Maximum capability (timing output)	
High level	-55 µA
Low level	9.6 mA
Driver output resistance	5Ω maximum
Timing output pulse width	1.45 µs maximum
Turn on time	220 ns maximum
Driver output pulse amplitude	3.2 V minimum, V_{CC} maximum

6.5 Module function. This module contains six line driver circuits and a 1.3 µs pulse width monostable multivibrator circuit. Each driver circuit is capable of driving 500 feet (152.3 meters) of type 2SWA or 3SWA cable, terminated in 75 ohms.

6.5.1 Typical driver circuit operation. Each driver circuit has a data input and a transformer coupled output. The monostable multivibrator circuit has two timing inputs and a timing output. To obtain an output pulse, a low-to-high transition must be applied to one of the timing inputs while the other timing input is held high. This produces a 1.3 µs pulse at any driver output which has its data input at a high level. This also produces the complement of the driver output pulse at the timing output. Signals presented to the data inputs must not change for 1.3 µs after the low-to-high transition applied to the timing input.

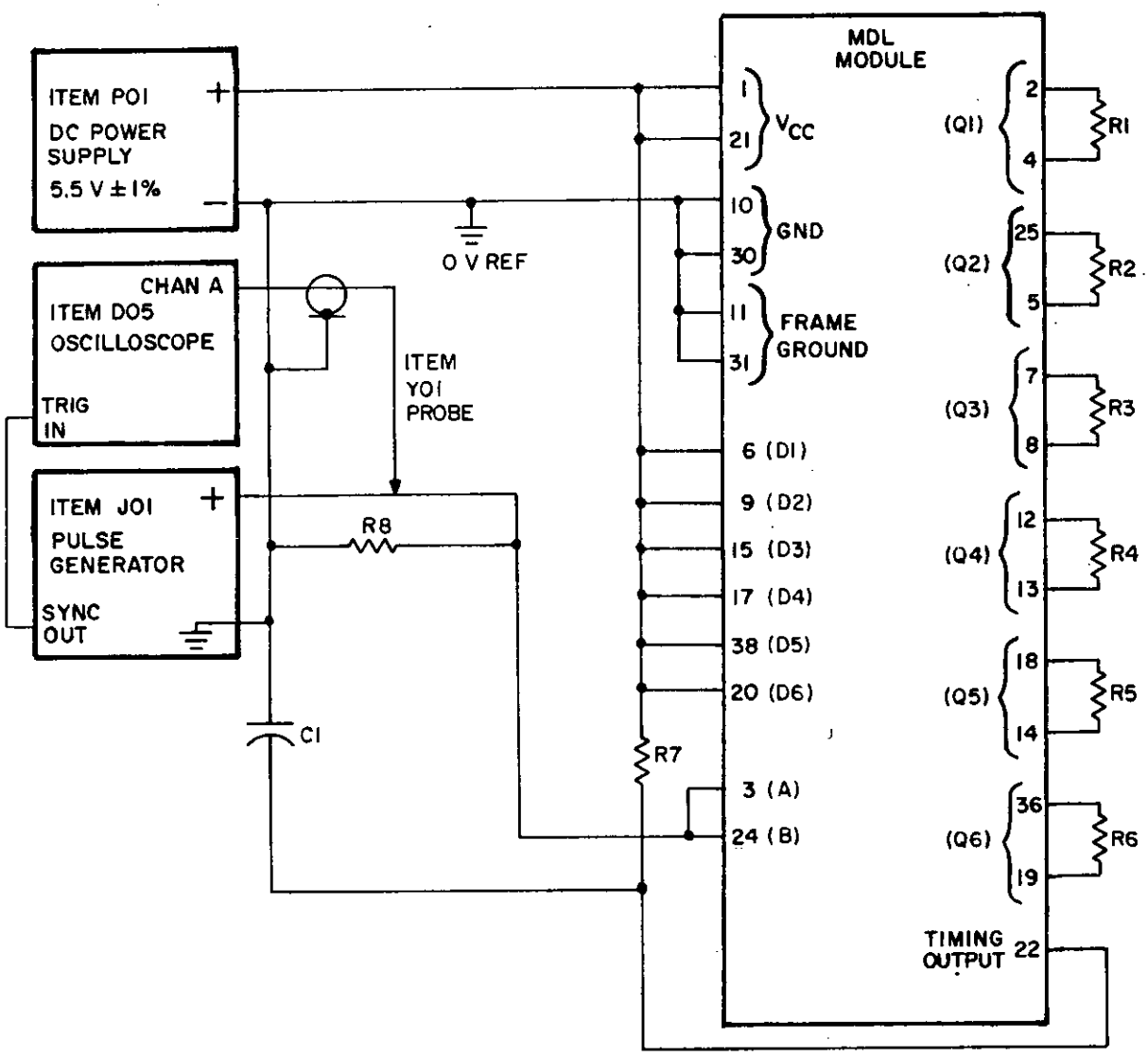
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N442)

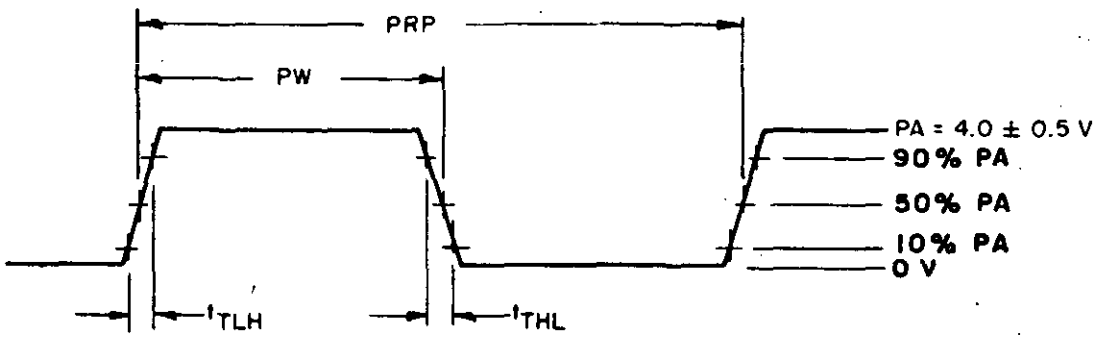
Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 511Ω ±1%, 1/4 W.
3. R8 = 50Ω ±5%, 1 W.
4. C1 = 220 pF ±10%, 15 V.
5. Delete capacitor C1 and connection between pins 1 and 10 for component temperature test only.
6. See figure 3 for input pulse parameters.

FIGURE 2. Operating temperature, vibration, component temperature,and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 3.0 ±0.5 µs.
PRP = 11 ±1 µs.
$t_{TLH} = t_{THL}$ = 20 ±10 ns.

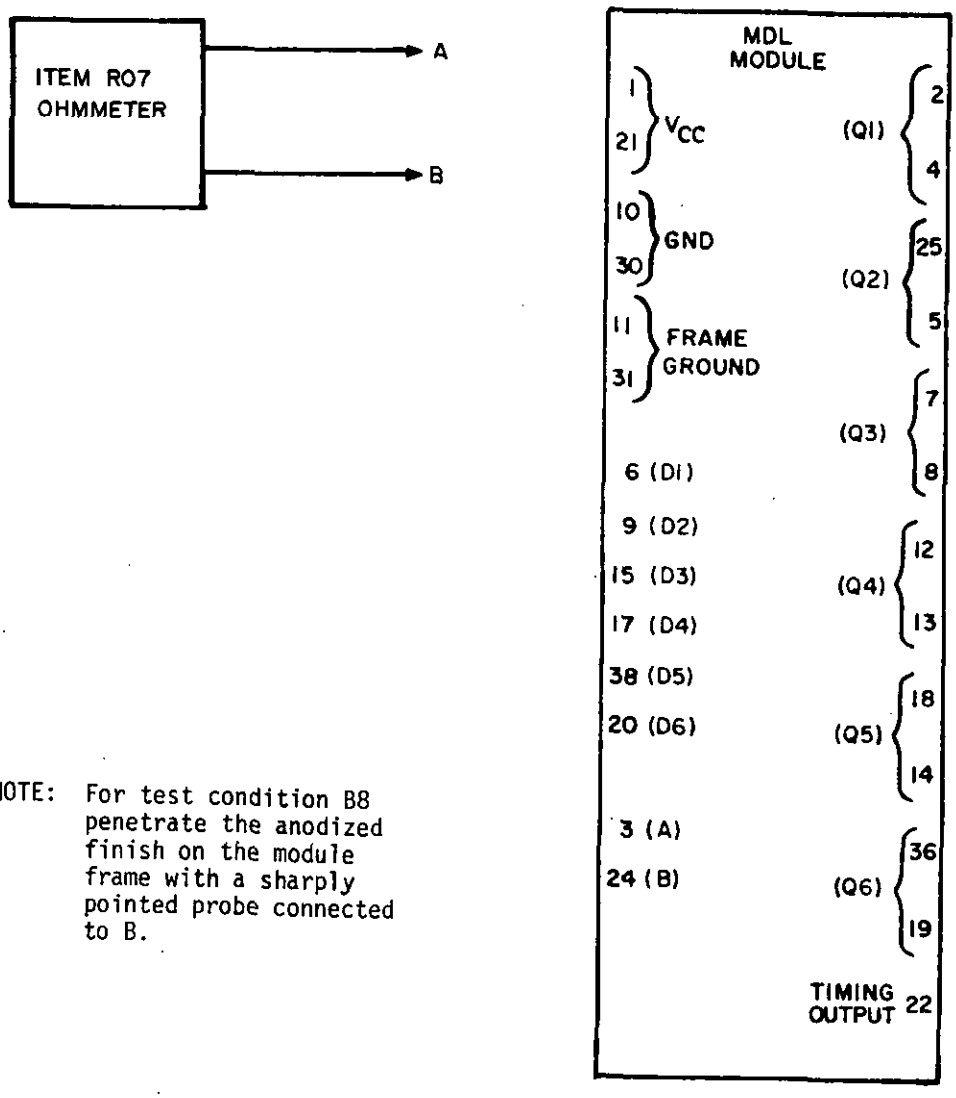
FIGURE 3. *Input pulse parameters for figure 2.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation and continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	11,31	1,2,3,4,5,6,7,8, 9,10,12,13,14,15, 16,17,18,19,20, 21,22,23,24,25, 26,27,28,29,30, 32,33,34,35,36, 37,38,39,40	10	∞	MΩ	4
A2	Isolation	16,23,26,27, 28,29,32,33, 34,35,37,39, 40	1,2,3,4,5,6,7,8, 9,10,11,12,13,14, 15,17,18,19,20, 21,22,24,25,30, 31,36,38	10	∞	MΩ	"
A3	Isolation	2,7,12,18,25, 36	1,10,21,30	10	∞	MΩ	"
B1	Continuity	11	31	0	5	Ω	"
B2	"	2	4	"	"	"	"
B3	"	25	5	"	"	"	"
B4	"	7	8	"	"	"	"
B5	"	12	13	"	"	"	"
B6	"	18	14	"	"	"	"
B7	"	36	19	"	"	"	"
B8	"	11	Frame	0	10	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: For test condition B8 penetrate the anodized finish on the module frame with a sharply pointed probe connected to B.

FIGURE 4. Isolation and continuity test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE III. Output noise voltage, timing output pulse amplitude, and driver output pulse amplitude test.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
C1	Q1 Output noise voltage	24	6,9, 15,17, 20,38	2	4	Open	3	0.0	0.3	V	5,6,8
C2	Q2 "	"	"	25	5	"	"	"	"	"	"
C3	Q3 "	"	"	7	8	"	"	"	"	"	"
C4	Q4 "	"	"	12	13	"	"	"	"	"	"
C5	Q5 "	"	"	18	14	"	"	"	"	"	"
C6	Q6 "	"	"	36	19	"	"	"	"	"	"
C7	Timing output pulse amplitude	"	"	22	11	"	"	3.8	5.5	"	5,6,9
C8	Timing output pulse amplitude	3	"	22	11	"	24	3.8	5.5	"	5,6,9
C9	Q1 Driver output pulse amplitude	9,15,17, 20,38	6,24	2	4	4	3	3.2	5.0	"	5,6,10
C10	Q2 "	6,15,17, 20,38	9,24	25	5	5	"	"	"	"	"
C11	Q3 "	6,9,17, 20,38	15,24	7	8	8	"	"	"	"	"
C12	Q4 "	6,9,15, 20,38	17,24	12	13	13	"	"	"	"	"
C13	Q5 Driver output pulse amplitude	6,9,15, 17,20	24,38	18	14	14	"	"	"	"	5,6,10
C14	Q6 Driver output pulse amplitude	6,9,15, 17,38	20,24	36	19	19	"	"	"	"	5,6,10
C15	Q1 Output noise voltage	6	9,15, 17,20, 24,38	2	4	Open	"	0.0	0.3	"	5,6,8
C16	Q2 "	9	6,15, 17,20 24,38	25	5	Open	"	0.0	0.3	"	5,6,8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE III. Output noise voltage, timing output pulse amplitude, and driver output pulse amplitude test - Continued.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
C17	Q3 Output noise voltage	15	6,9, 17,20, 24,38	7	8	Open	3	0.0	0.3	V	5,6,8
C18	Q4 "	17	6,9, 15,20, 24,38	12	13	"	"	"	"	"	"
C19	Q5 "	38	6,9, 15,17, 20,24	18	14	"	"	"	"	"	"
C20	Q6 "	20	6,9, 15,17, 24,38	36	19	"	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE IV. Output pulse characteristics.

Test condition	Parameter	Connect						Initial limits 25°C±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
E1	Q1 Output pulse characteristics										
	Pulse amplitude	Open	6,9,15,17,	2	4	4	3	3.2	5.5	V	5,6,10
	Pulse width	"	20,24,38	"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	52	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	10	70	ns	"
	Droop	"	"	"	"	"	"	-0.3	0.3	V	"
	Undershoot	"	"	"	"	"	"	0.0	2.0	V	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"
E2	Q2 Output pulse characteristics										
	Pulse amplitude	"	"	25	5	5	"	3.2	5.5	V	"
	Pulse width	"	"	"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	52	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	10	70	ns	"
	Droop	"	"	"	"	"	"	-0.3	0.3	V	"
	Undershoot	"	"	"	"	"	"	0.0	2.0	V	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"
E3	Q3 Output pulse characteristics										
	Pulse amplitude	"	"	7	8	8	"	3.2	5.5	V	"
	Pulse width	"	"	"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	52	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	10	70	ns	"
	Droop	"	"	"	"	"	"	-0.3	0.3	V	"
	Undershoot	"	"	"	"	"	"	0.0	2.0	V	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"
E4	Q4 Output pulse characteristics										
	Pulse amplitude	"	"	12	13	13	"	3.2	5.5	V	"
	Pulse width	"	"	"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	52	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	10	70	ns	"
	Droop	"	"	"	"	"	"	-0.3	0.3	V	"
	Undershoot	"	"	"	"	"	"	0.0	2.0	V	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE IV. Output pulse characteristics - Continued.

Test condition	Parameter	Connect						Initial limits 25°C±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
E5	Q5 Output pulse characteristics										
	Pulse amplitude	Open	6,9,15,17,	18	14	14	3	3.2	5.5	V	5,6,10
	Pulse width	"	20,24,38	"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	52	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	10	70	ns	"
	Droop	"	"	"	"	"	"	-0.3	0.3	V	"
	Undershoot	"	"	"	"	"	"	0.0	2.0	V	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"
E6	Q6 Output pulse characteristics										
	Pulse amplitude	"	"	36	19	19	"	3.2	5.5	V	"
	Pulse width	"	"	"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	52	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	10	70	ns	"
	Droop	"	"	"	"	"	"	-0.3	0.3	V	"
	Undershoot	"	"	"	"	"	"	0.0	2.0	V	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"
E7	Timing output pulse characteristics										
	Pulse amplitude	"	"	22	11	Open	"	3.8	5.5	V	5,6,11
	Pulse width	"	"	"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	30	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	100	350	ns	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"
E8	Timing output pulse characteristics										
	Pulse amplitude	"	3,6,9,15,	"	"	"	24	3.8	5.5	V	"
	Pulse width	"	17,20,38	"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	30	ns	"

Source: https://assist.dla.mil -- Downloaded: 20260[illegible]-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE IV. Output pulse characteristics - Continued.

Test condition	Parameter	Connect						Initial limits 25°C±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
	Transition time (t_{THL})	Open	3,6,9,15,17, 20,38	22	11	Open	24	100	350	ns	5,6,11
	Turn on time	"	"	"	"	"	"	5	220	ns	"
E9	Timing output pulse characteristics, V_{CE} (SAT)	"	"	"	"	"	"	0.00	0.45	V	"
F1	Q1 Output pulse characteristics										
	Pulse amplitude	9,15,17, 20,38	6,24	2	4	4	3	3.5	5.5	V	5,7,10
	Pulse width	"		"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	---	---	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	---	---	ns	"
	Droop	"	"	"	"	"	"	0.0	0.3	V	"
	Undershoot	"	"	"	"	"	"	0.0	2.0	V	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"
F2	Timing output pulse characteristics										
	Pulse amplitude	Open	24	22	11	Open	"	3.8	5.5	V	5,7,11
	Pulse width	"		"	"	"	"	1.10	1.45	µs	"
	Transition time (t_{TLH})	"	"	"	"	"	"	5	30	ns	"
	Transition time (t_{THL})	"	"	"	"	"	"	100	350	ns	"
	Droop	"	"	"	"	"	"	---	---	V	"
	Undershoot	"	"	"	"	"	"	---	---	V	"
	Turn on time	"	"	"	"	"	"	5	220	ns	"
F3	Timing output pulse characteristics V_{CE} (SAT)	"	"	"	"	"	"	0.00	0.45	V	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.

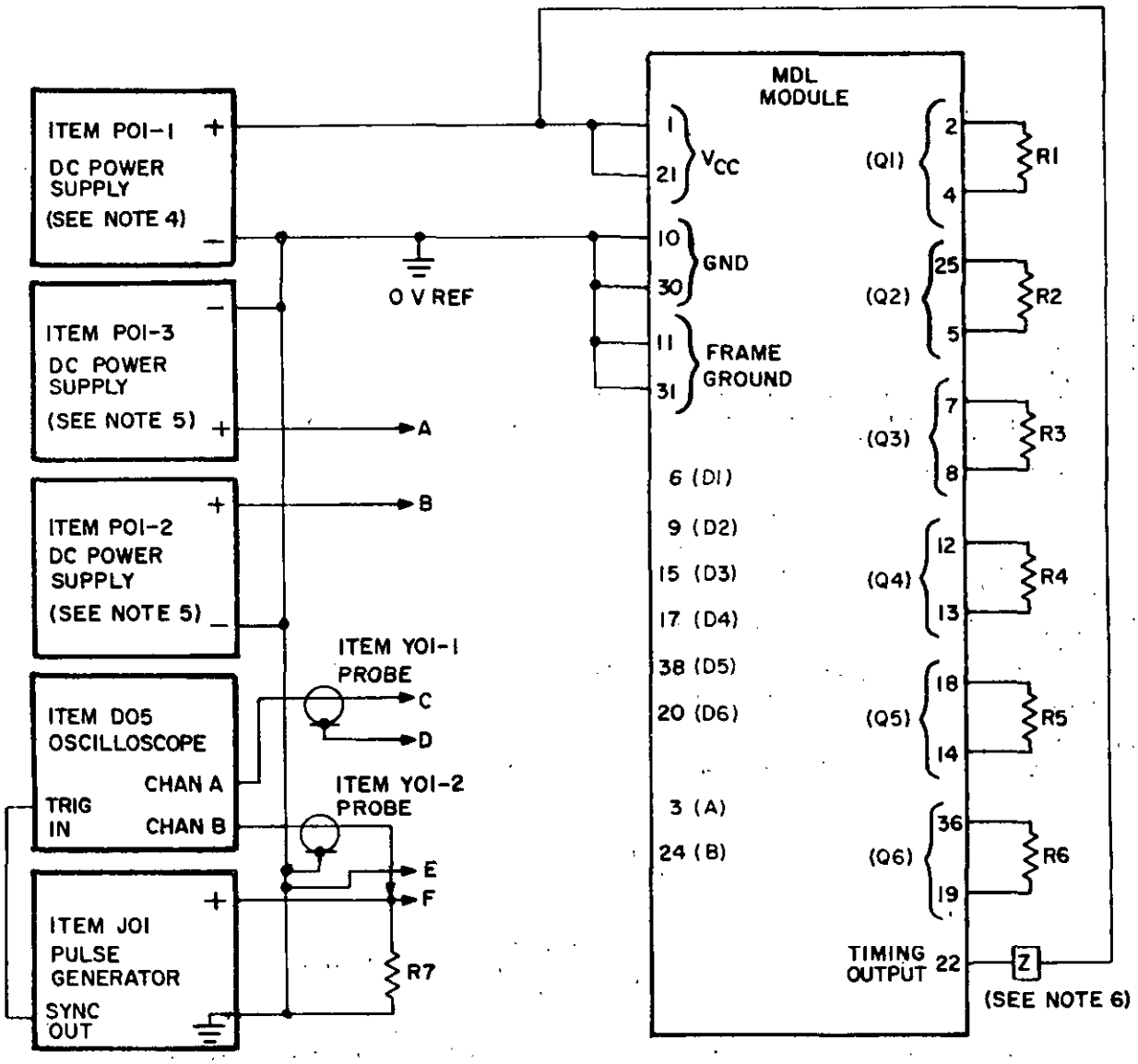


FIGURE 5. Output noise voltage, timing output pulse amplitude, driver output pulse amplitude, and output pulse characteristics test circuit for tables III and IV.

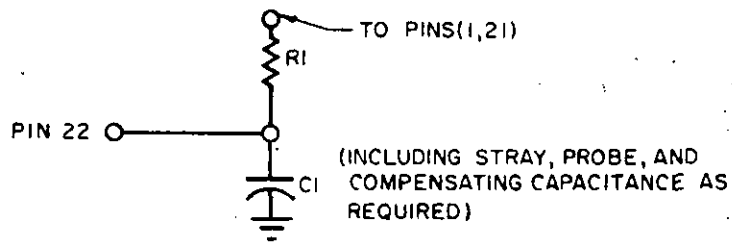
Source: https://assist.dla.mil Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 50Ω ±5%, 1 W.
3. See figures 6 and 7 for input pulse parameters.
4. For test conditions C1-C20 set item P01-1 at 4.5 V ±1%. For test conditions E1-E9 and F1-F3 set item P01-1 at 5.5 V ±1%.
5. Set items P01-2 and P01-3 as follows:

Item	Operating temperature			
	0°C	25°C	60°C	80°C
P01-2 ±1%	2.0 V	1.9 V	1.8 V	1.8 V
P01-3 ±1%	1.2 V	1.1 V	0.95 V	0.8 V

6. The impedance Z is composed of resistor R1 from pin 22 to pin 1 and a 30 pF ±5%, 15 V, capacitor C1 from pin 22 to ground as specified in note 8.

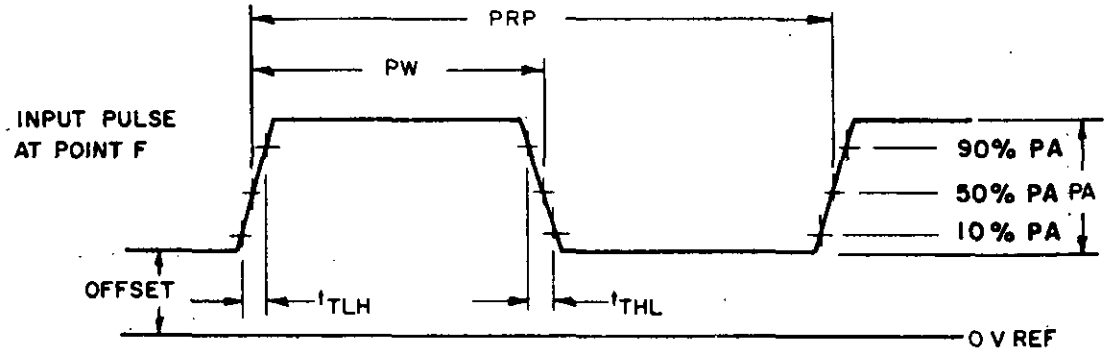
   R1 = 250Ω ±1%, 1/4 W, for test condition E9.
   R1 = 3.83 kΩ ±1%, 1/8 W, for all other test conditions.



7. The loop length from pin 1 to pin 22, including the length of the body of the resistor, shall not exceed 4.5 inches (114.3 mm).
8. With the module removed from the test connector, measure the capacitance between test connector pin 22 and pins 10, 11, 30, and 31 tied together. The measurements shall be performed at the point of mechanical contact between module blade pin and the test connector pins. The measured value at 1 MHz ±10%, plus probe capacitance, shall be 30 ±5 pF. For these measurements disconnect resistor R1 from pin 22. The resistor should be disconnected as close to its body as practical. Reconnect the resistor prior to using the test connector for module testing.

FIGURE 5. Output noise voltage, timing output pulse amplitude, driver output pulse amplitude, and output pulse characteristics test circuit for tables III and IV - Continued.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.

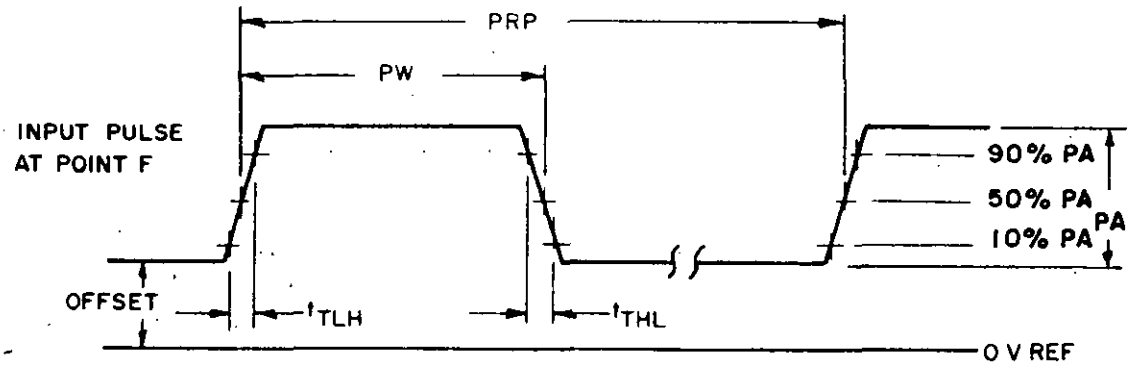


NOTES:
1. Input pulse parameters for test conditions C1-C20, E1-E9, M1 and M2:
   PW = 3 µs ±10%.
   PRP = 6 µs ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
2. PA and offset for module operating temperatures of 0°C, 25°C, 60°C, and 80°C.

Parameter	Operating temperature			
	0°C	25°C	60°C	80°C
PA ±1%	0.8 V	0.8 V	0.85 V	1.0 V
Offset ±1%	1.2 V	1.1 V	0.95 V	0.8 V

FIGURE 6. Input pulse parameters for test conditions C1-C20, E1-E9, M1 and M2, tables III, IV, and VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters for test conditions F1-F3:
   PW = 100 ns ±10%.
   PRP = 1 ms ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
2. PA and offset for module operating temperatures of 0°C, 25°C, and 60°C.

Parameter	Operating temperature		
	0°C	25°C	60°C
PA ±1%	0.8 V	0.8 V	0.85 V
Offset ±1%	1.2 V	1.1 V	0.95 V

FIGURE 7. Input pulse parameters for test conditions F1-F3, table IV.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.



FIGURE 8. Output noise voltage test circuit waveform for test conditions C1-C6 and C15-C20, table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

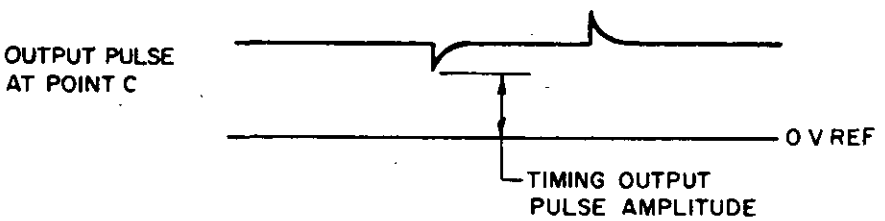
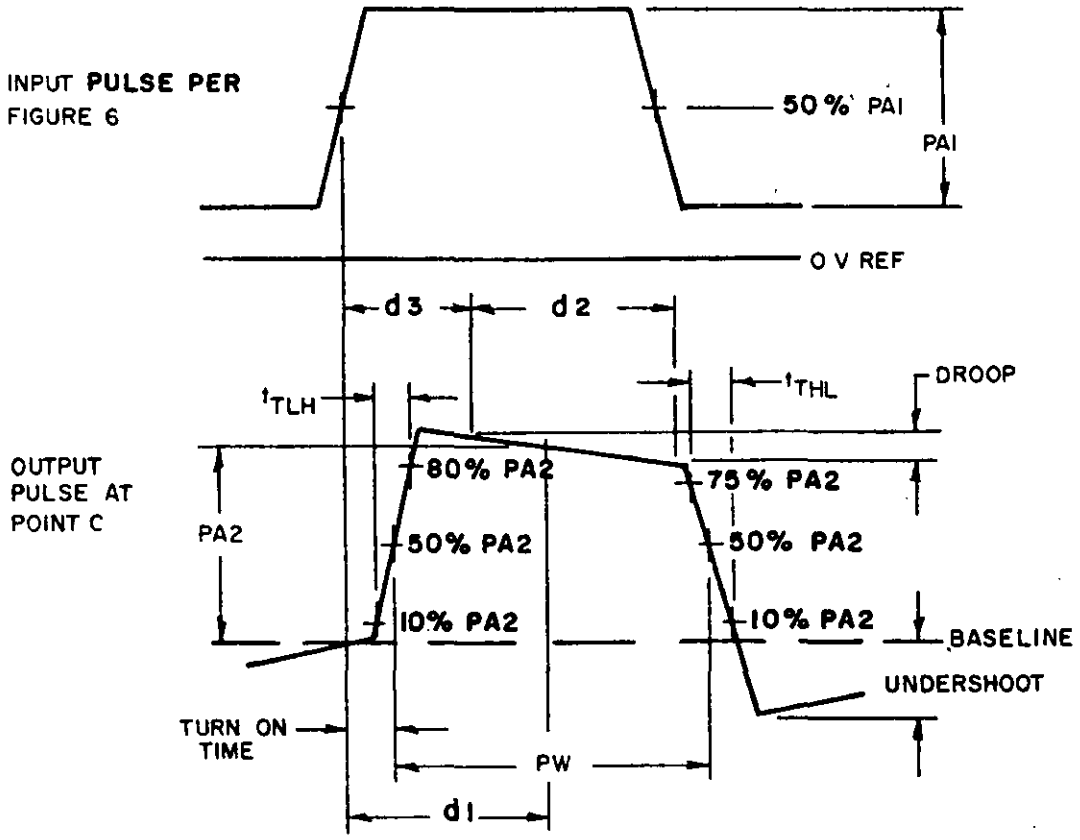


FIGURE 9. Timing output pulse amplitude test circuit waveform for test conditions C7 and C8, table III.

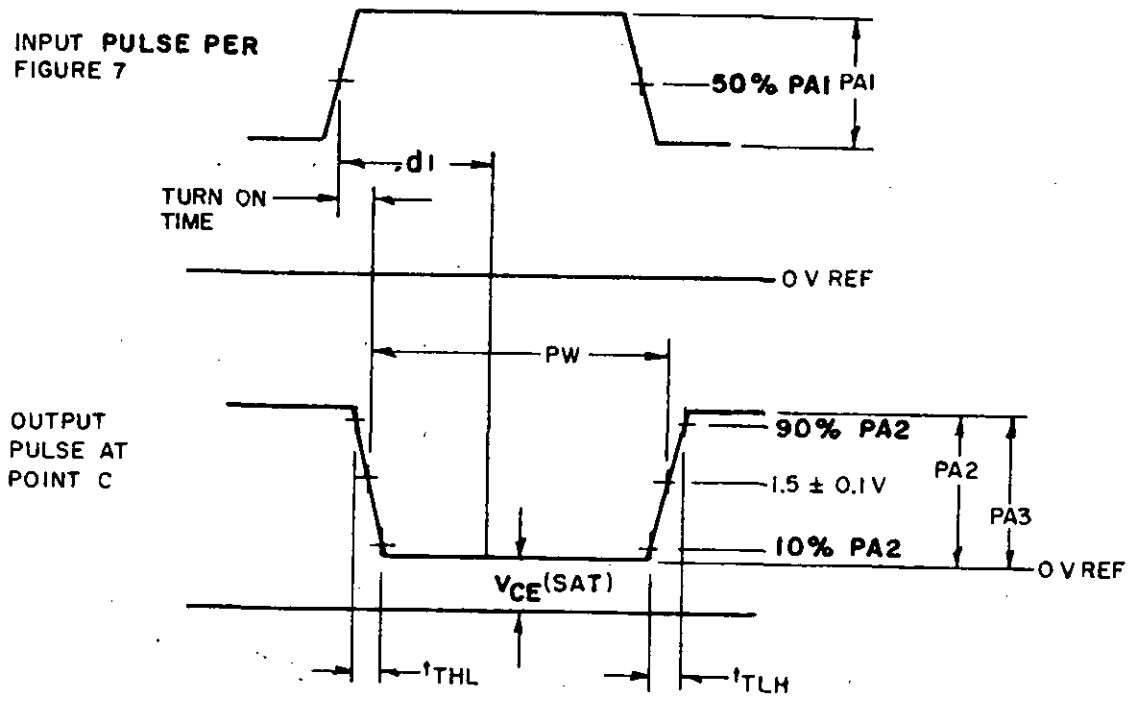
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Pulse amplitude and droop are measured at the specified delays:
d1 = 600 ns.
d2 = 700 ns.
d3 = 300 ns.

FIGURE 10. Driver output pulse amplitude (test conditions C9-C14, table III) and output pulse characteristics for test conditions E1-E6 and F1, table IV.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: V_{CE}(SAT) is measured at the specified delay:
d1 = 700 ns.

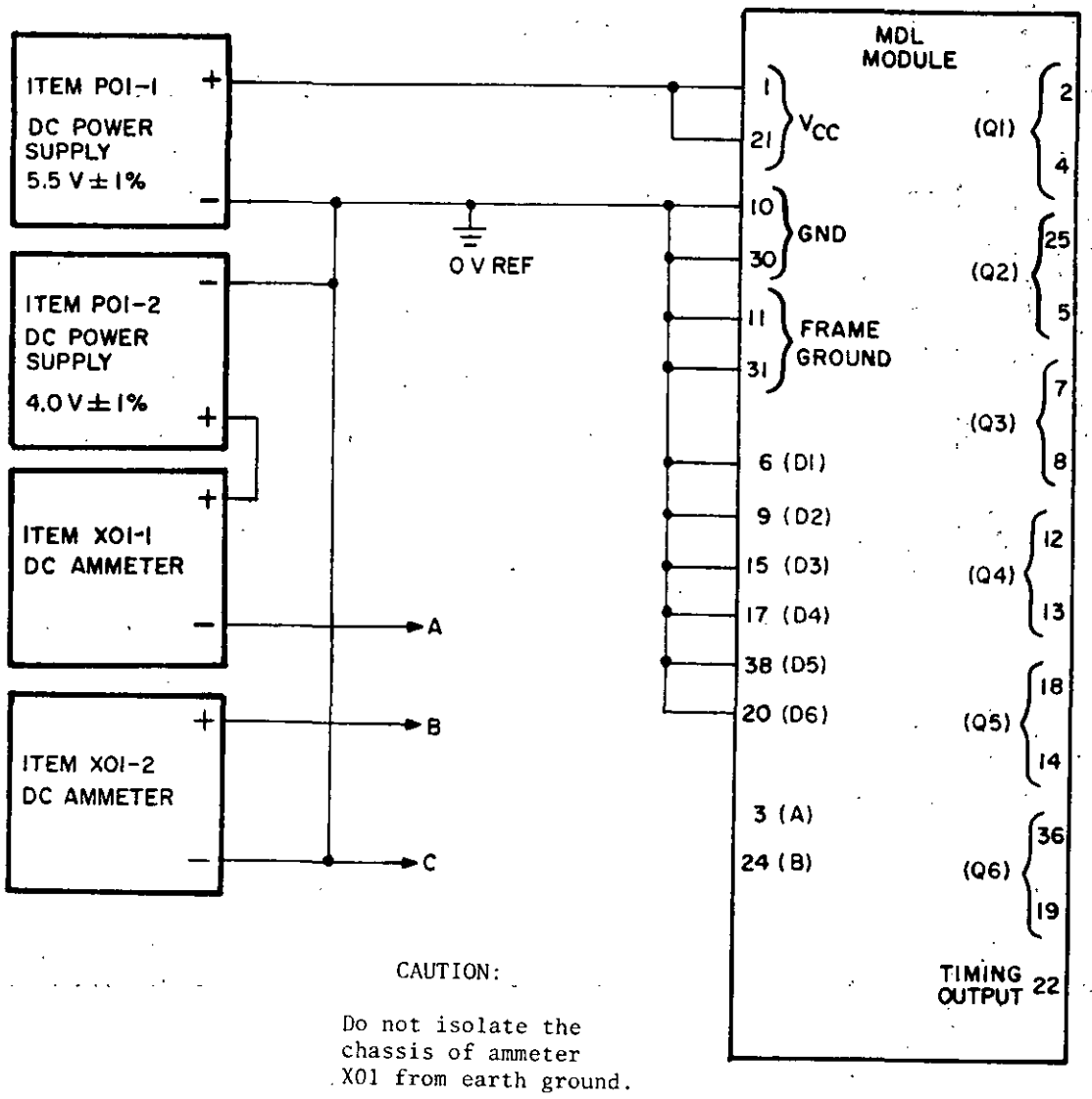
FIGURE 11. Timing output pulse characteristics test circuit waveforms for test conditions E7-E9, F2 and F3, table IV.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE V. Input current (timing, low-level and high level), input current (data, high-level), input voltage (data, low-level), and output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Input current, timing, low level	Open	24	Open	0.0	1.6	mA	12
G2	Input current, timing, low level	Open	3	Open	"	1.6	mA	"
H1	Input current, timing, high level	3	Open	24	"	2.0	µA	"
H2	Input current, timing, high level	24	Open	3	"	2.0	"	"
J1	Input current, data, high level	6	Not used	Not used	"	4.0	"	13
J2	"	9	"	"	"	"	"	"
J3	"	15	"	"	"	"	"	"
J4	"	17	"	"	"	"	"	"
J5	"	38	"	"	"	"	"	"
J6	"	20	"	"	"	"	"	"
K1	Input voltage, data, low level	6	"	"	0.00	0.32	V	14,15
K2	"	9	"	"	"	"	"	"
K3	"	15	"	"	"	"	"	"
K4	"	17	"	"	"	"	"	"
K5	"	38	"	"	"	"	"	"
K6	"	20	"	"	"	"	"	"
L	Output short circuit current	22	"	"	0.7	3.0	mA	16

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]-20T16:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Use item X01-2 for test conditions G1 and G2 and item X01-1 for test conditions H1 and H2.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 12. Input current (timing, low-level and high-level) test circuit for test conditions G1, G2, H1, and H2, table V.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.

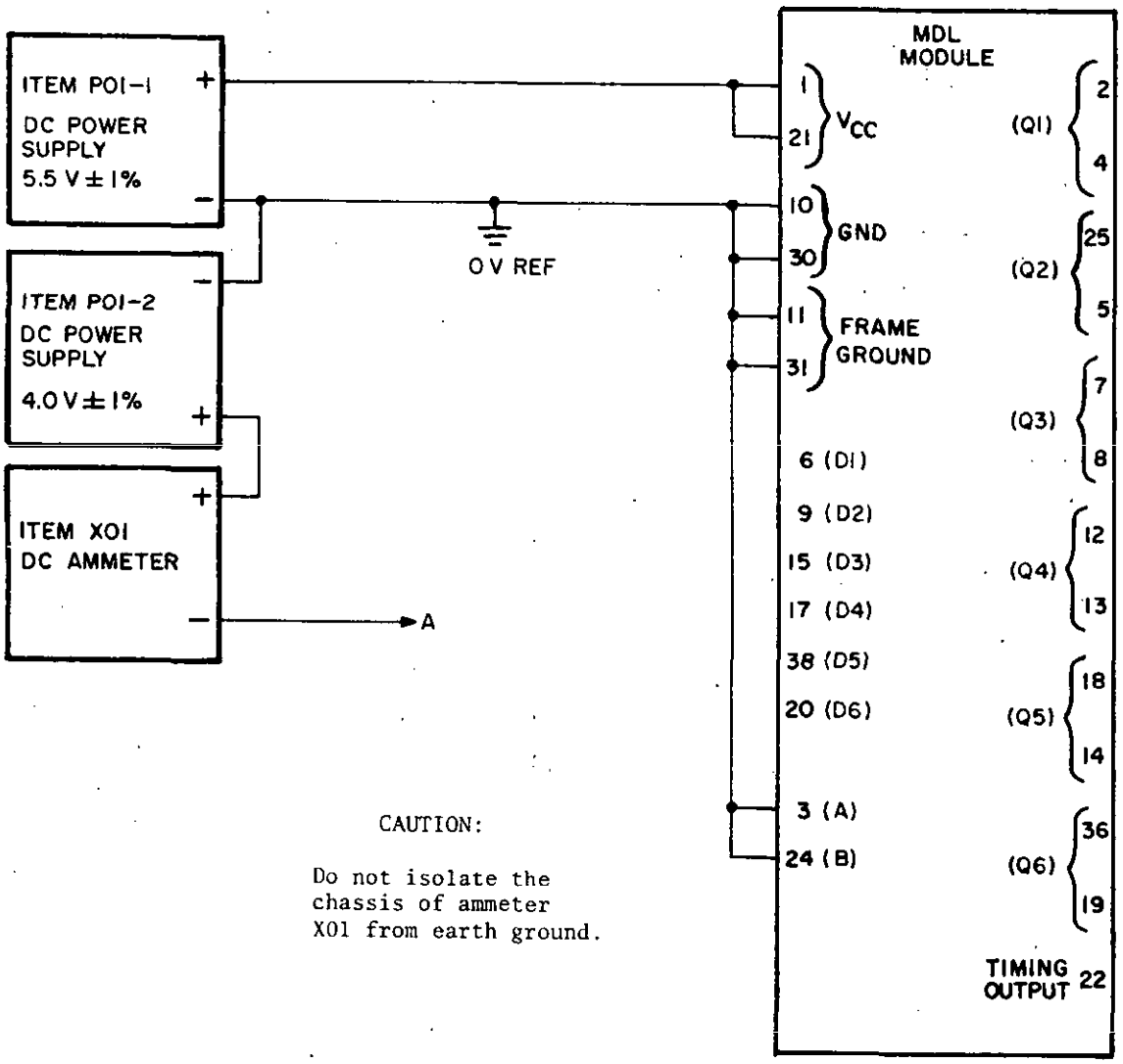
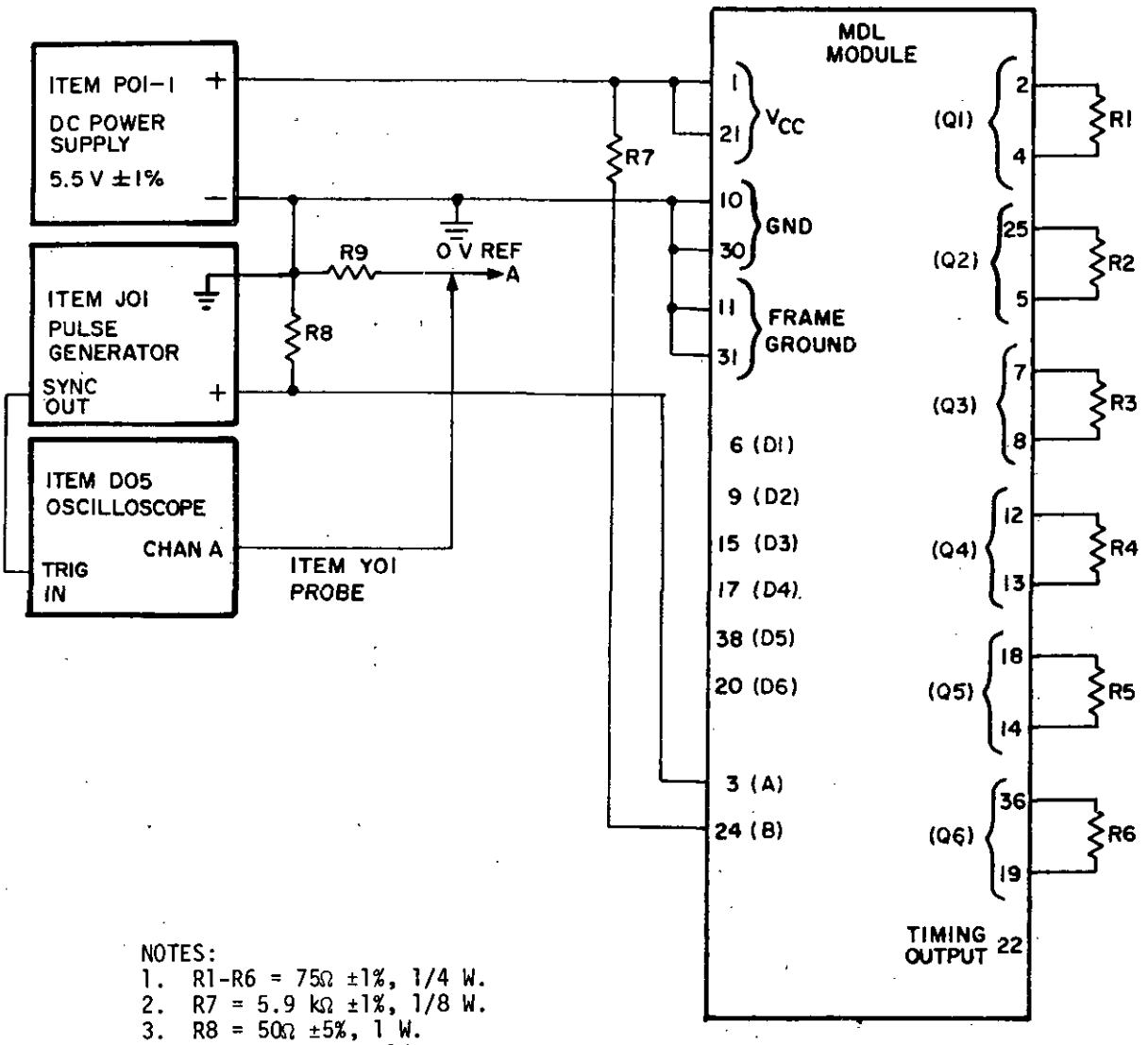


FIGURE 13. Input current (data, high-level) test circuit for test conditions J1-J6, table V.

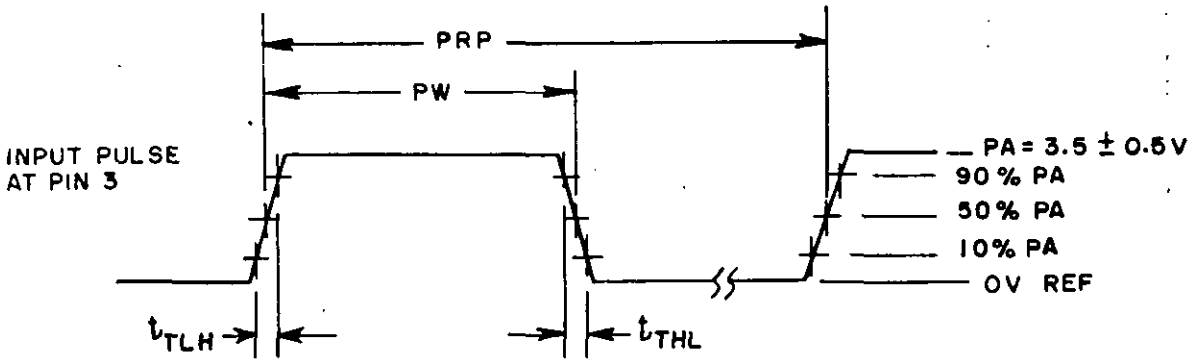
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 5.9 kΩ ±1%, 1/8 W.
3. R8 = 50Ω ±5%, 1 W.
4. R9 = 200Ω ±1%, 1/4 W.

FIGURE 14. Input voltage (data, low-level) test circuit for test conditions K1-K6, table V.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 3.0 ±0.5 µs.
PRP = 15 ±3 µs.
$t_{TLH} = t_{THL}$ = 50 ±20 *ns*.



FIGURE 15. Input voltage (data, low-level) test circuit waveforms for test conditions K1-K6, table V.

Source: https://assist.dla.mil Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

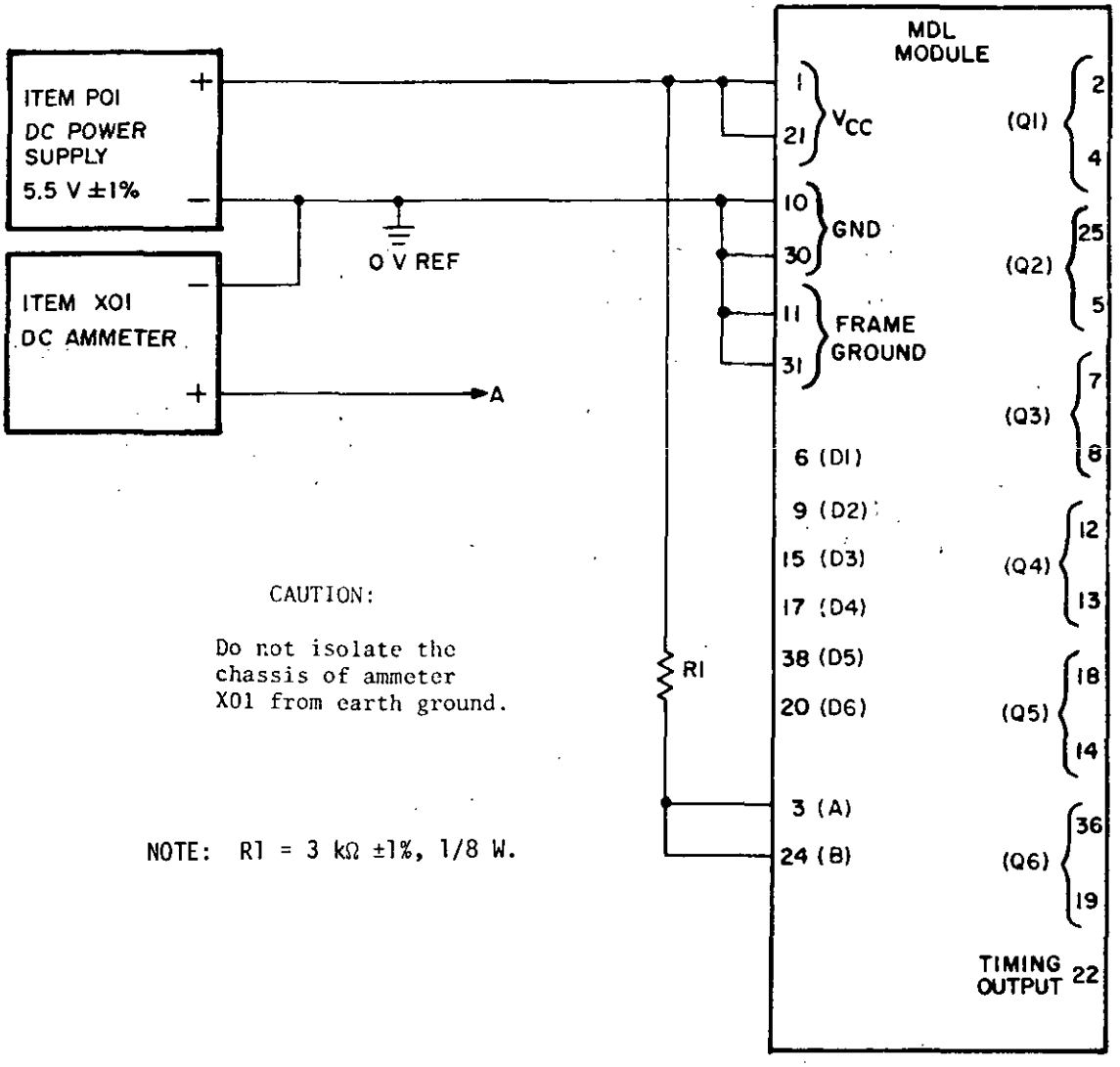


FIGURE 16. Output short circuit current test circuit for test condition L, table V.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE VI. Power supply test.

Test condition	Parameter	Connect			Initial limits 25C° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Power supply current	Open	3,6,9,15, 17,20,24, 38	Open	0	50	mA	6,17
M2	Power supply current	3	Open	6,9,15,17, 20,24,38	0	150	mA	6,17

TABLE VII. Zener applied potential test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
N	Zener applied potential	0	150	mA	18,19

Source: https://assist.dla.mil -- Downloaded: 20260420T16:03Z
Check the source to verify that this is the current version before use.

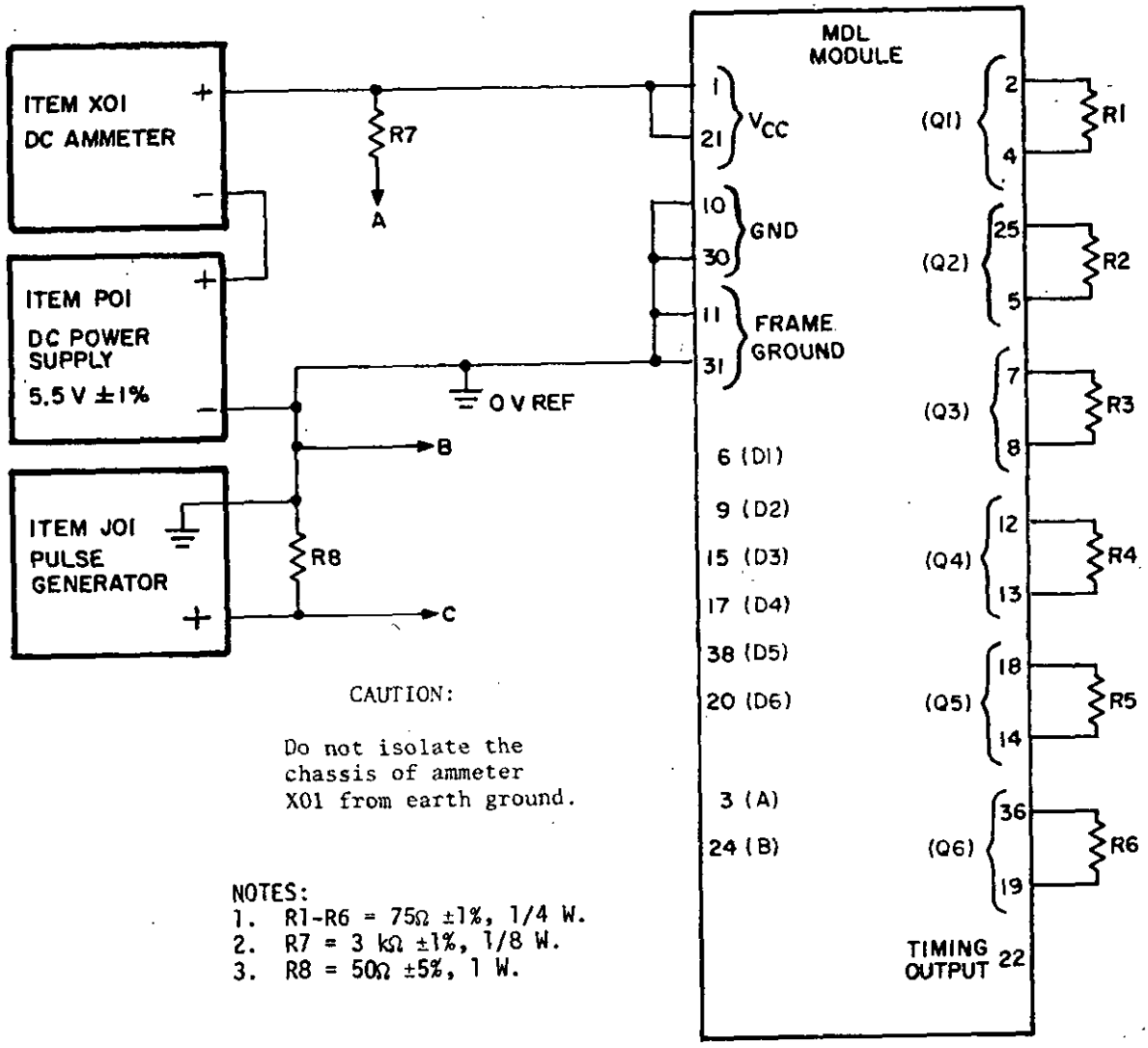


FIGURE 17. Power supply current test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.

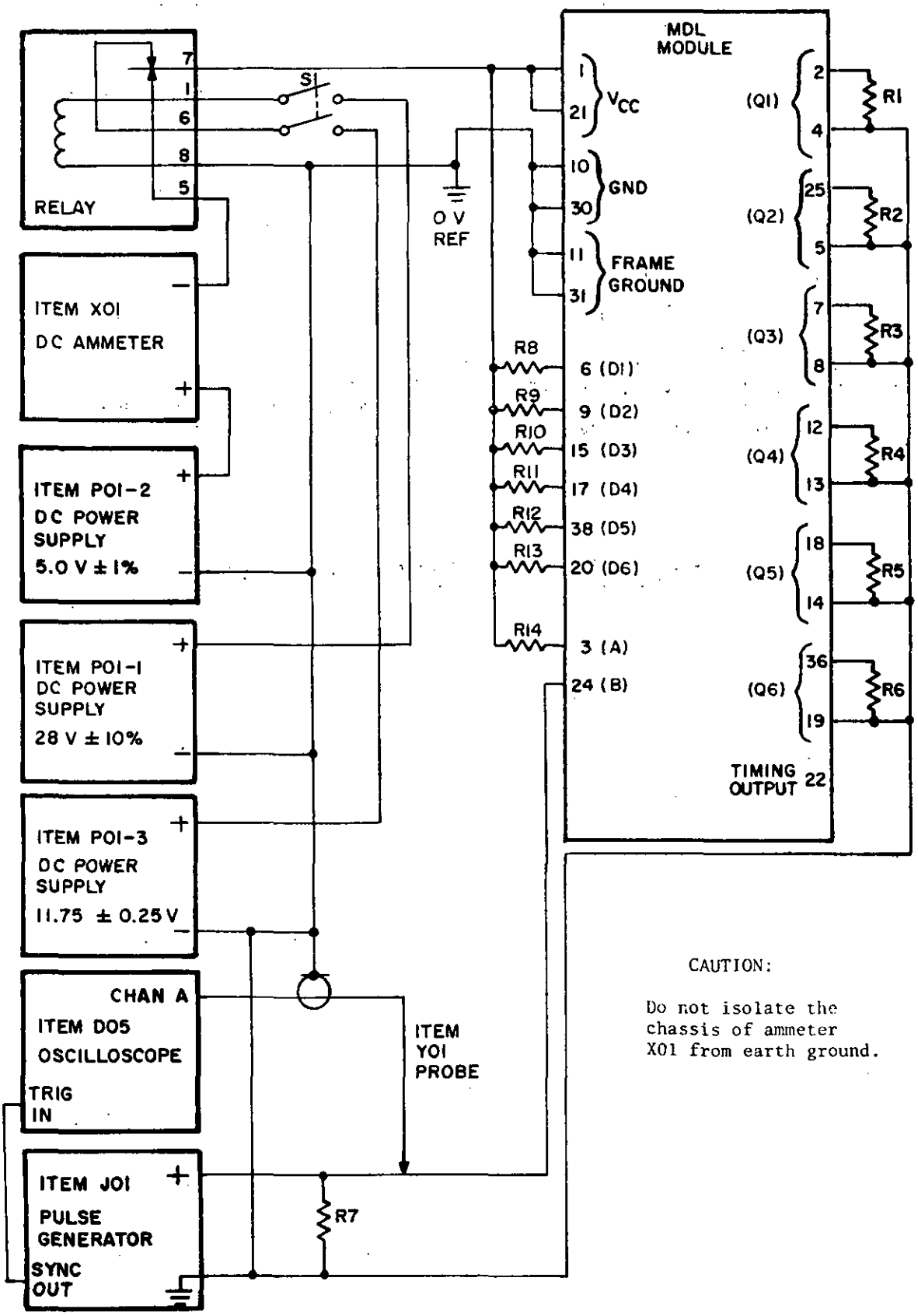


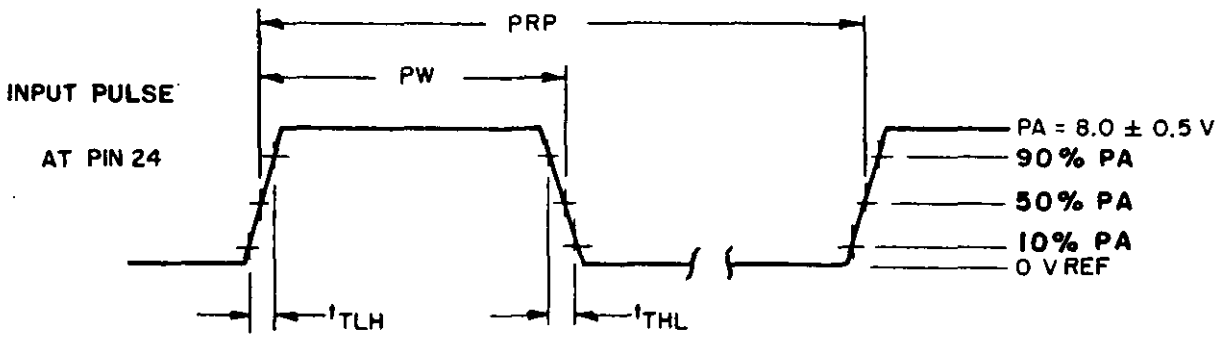
FIGURE 18. Zener applied potential test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 50Ω ±5%, 1 W.
3. R8-R14 = 1 kΩ ±1%, 1/8 W.
4. S1 = DPST switch.
5. Relay = Agastat Model 2112, or equivalent.
6. The relay shall be adjusted for a 1.0 ±0.2 second delay.
7. Close switch S1 and after 1.2 seconds read item X01. At the conclusion of the test switch S1 shall be opened.

FIGURE 18. Zener applied potential test circuit for table VII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 3.0 ±0.5 µs.
PRP = 15 ±3 µs.
$t_{TLH} = t_{THL}$ = 100 ±50 ns.

FIGURE 19. Zener applied potential test circuit waveform for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

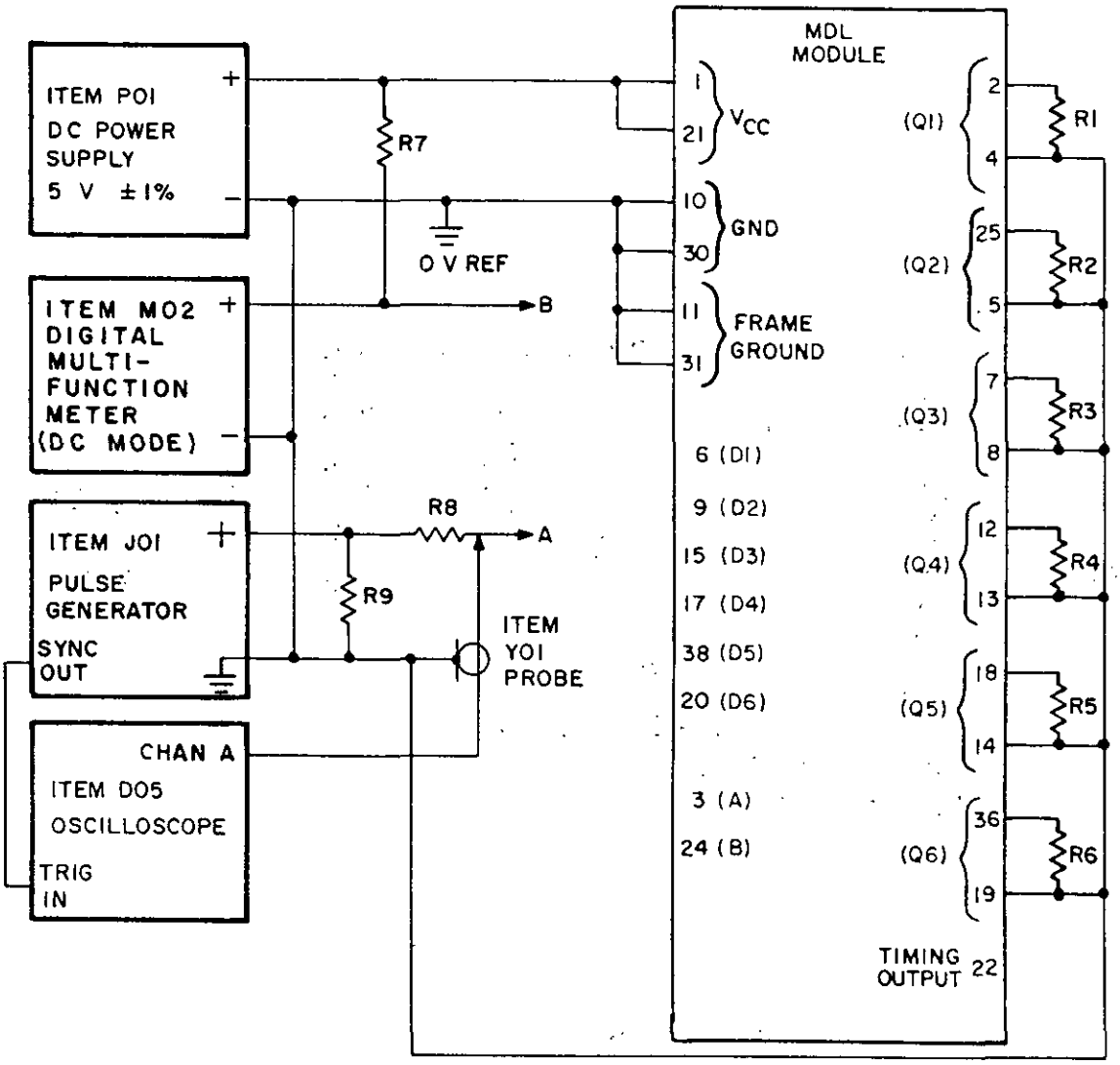
TABLE VIII. Reverse voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Reverse voltage	6	6	3.8	5.5	V	20,21
P2	"	9	9	"	"	"	"
P3	"	15	15	"	"	"	"
P4	"	17	17	"	"	"	"
P5	"	38	38	"	"	"	"
P6	"	20	20	"	"	"	"
P7	"	3	3	"	"	"	"
P8	"	24	24	"	"	"	"

TABLE IX. V_{OL} loaded output test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
Q	V_{OL} loaded output	0.0	1.1	V	22,23

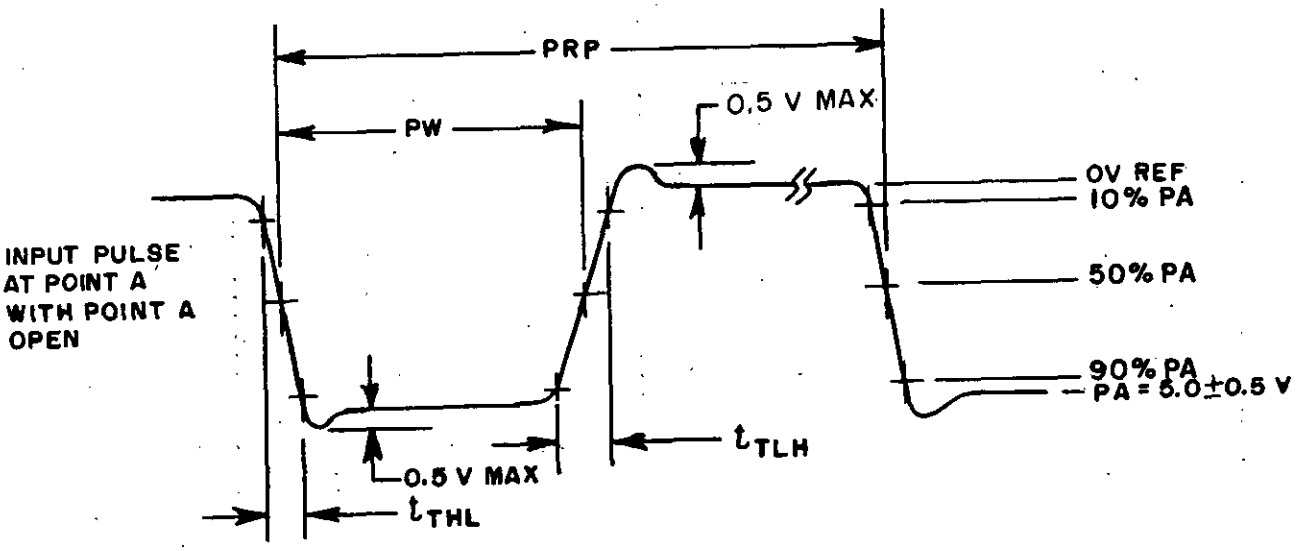
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 5.9 kΩ ±1%, 1/8 W.
3. R8 = 127Ω ±1%, 1/8 W.
4. R9 = 50Ω ±5%, 1 W.
5. Before connecting point B, apply point A to the pin specified in table VIII for 10 seconds minimum.
6. Disconnect point A and connect point B and measure voltage with item M02.

* FIGURE 20. Reverse voltage test circuit for table VIII.

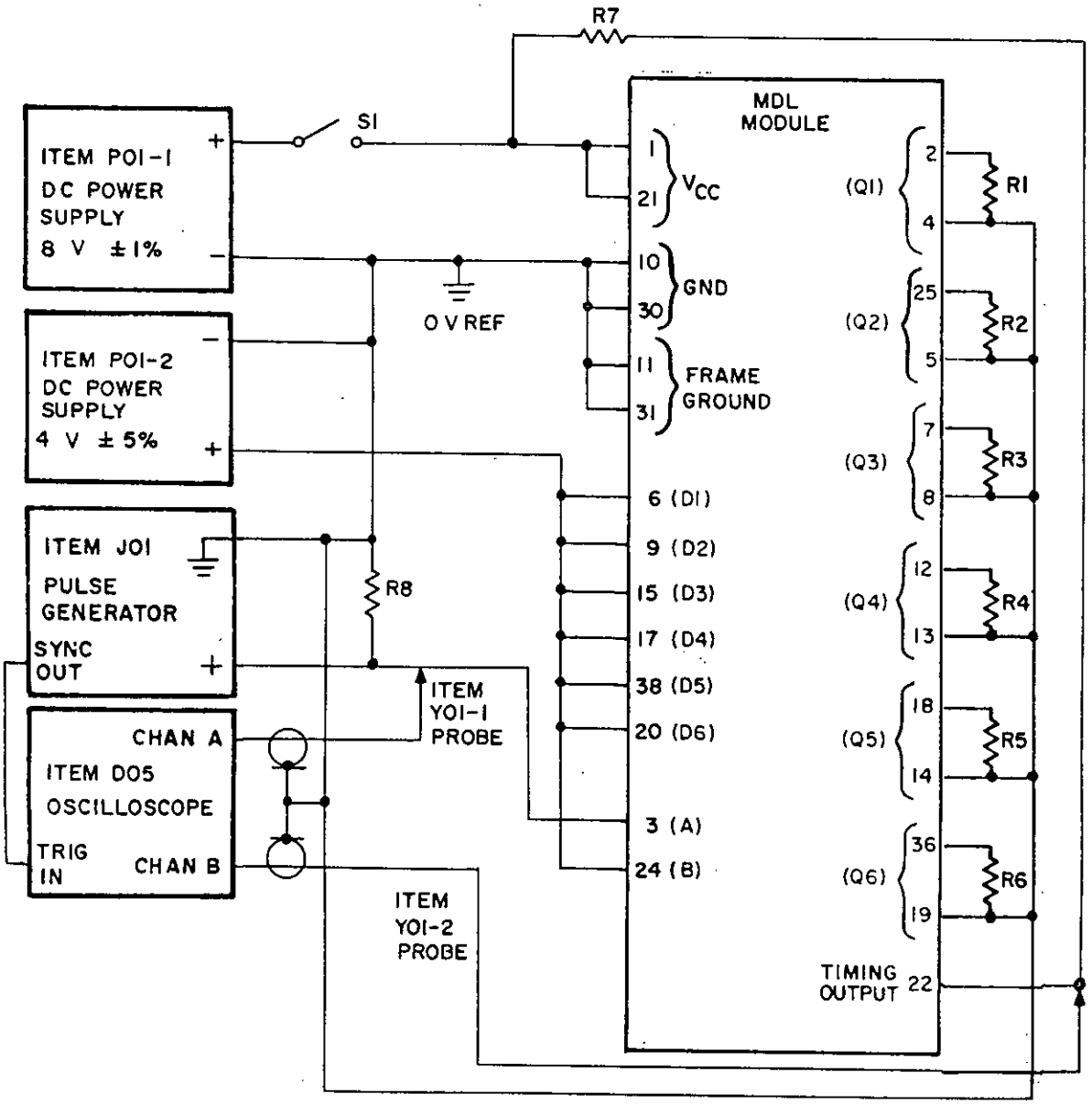
Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 10 ±1 µs.
PRP = 33 ±3 µs.
$t_{TLH} = t_{THL}$ = 100 ±20 ns.

FIGURE 21. Reverse voltage test circuit waveform for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

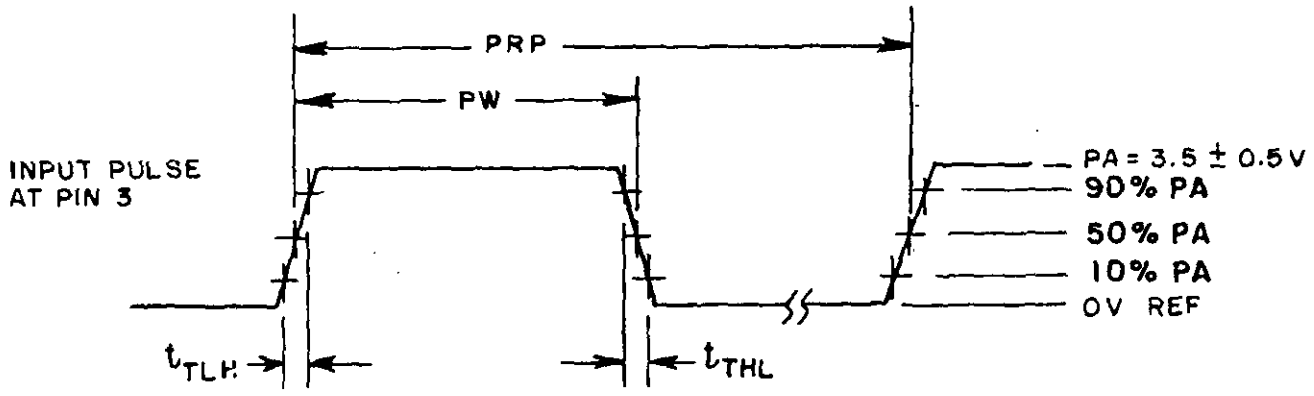


NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 511Ω ±1%, 1/4 W.
3. R8 = 50Ω ±5%, 1 W.
4. S1 = SPST switch.
5. Test sequence:

   a. Close switch S1 for a minimum of 30 seconds.
   b. Measure $V_{CE}(SAT)$ on channel B of item DO5.

FIGURE 22. V_{OL} loaded output test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 3.0 ±0.5 µs.
PRP = 15 ±3 µs.
$t_{TLH} = t_{THL}$ = 50 ±20 ns.

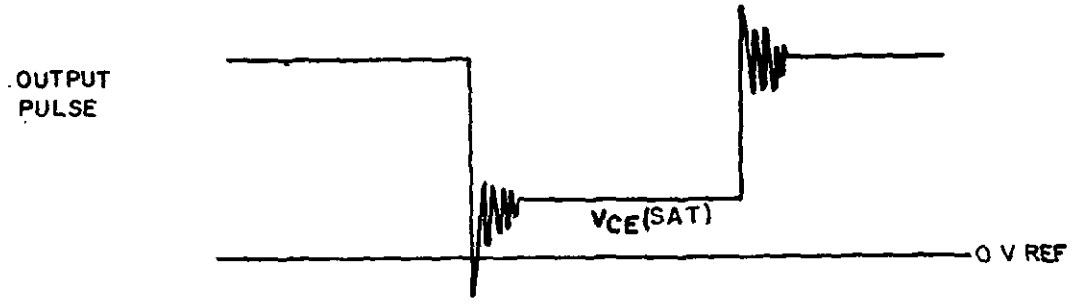


FIGURE 23. V_{OL} loaded output test circuit waveforms for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

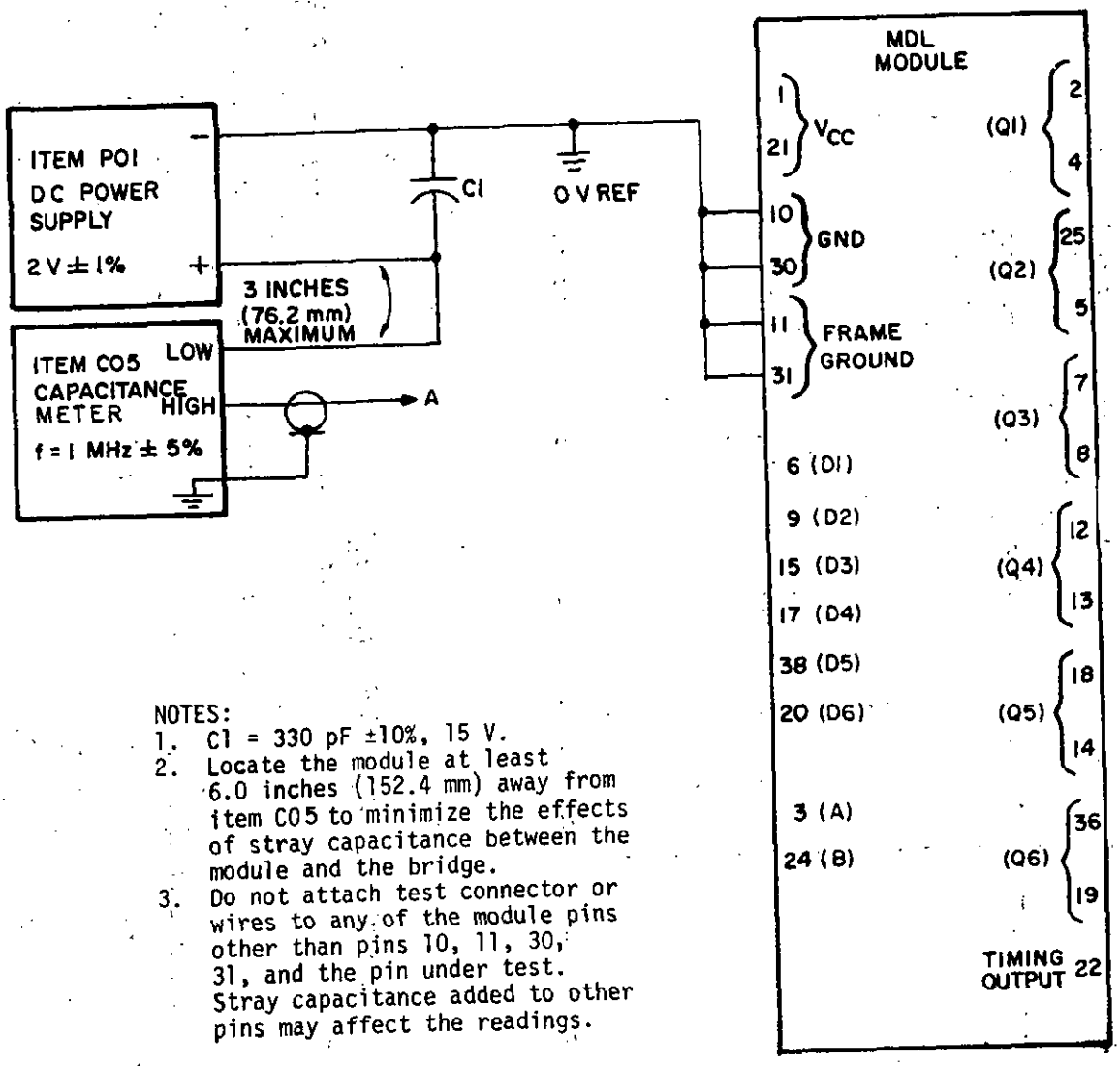
TABLE X. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
R1	Input capacitance	6	0	20	pF	24
R2	"	9	"	"	"	"
R3	"	15	"	"	"	"
R4	"	17	"	"	"	"
R5	"	38	"	"	"	"
R6	"	20	"	"	"	"
R7	"	3	"	"	"	"
R8	"	24	"	"	"	"

TABLE XI. Input transient immunity test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Input transient immunity	22	Open	24	Open	2.0	5.5	V	25,26
S2	Input transient immunity	22	Open	3	Open	2.0	5.5	V	25,26
S3	Input transient immunity	D	6,9,15,17, 20,38	3,24	A	0	300	ns	25,26

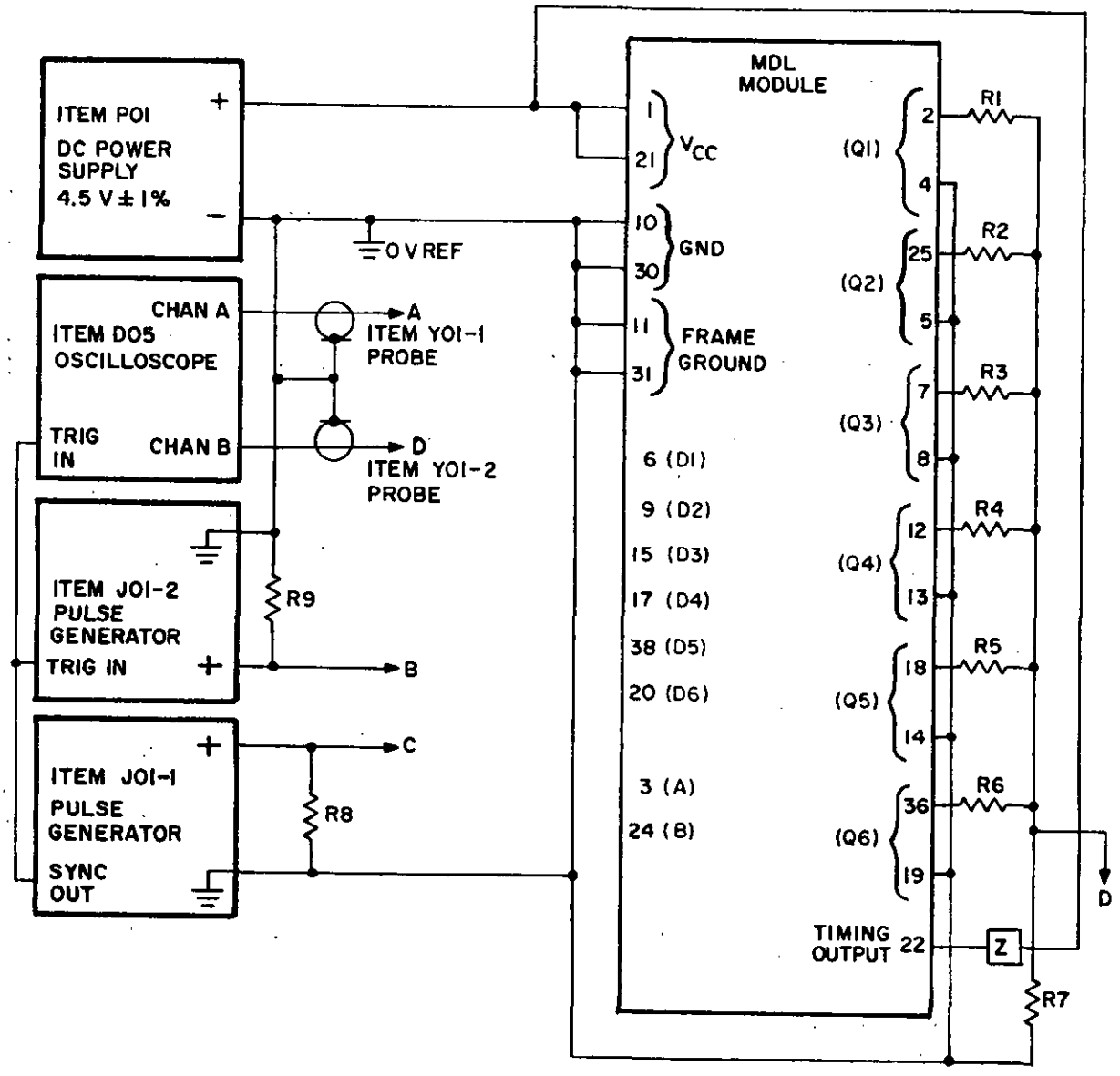
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. C1 = 330 pF ±10%, 15 V.
2. Locate the module at least 6.0 inches (152.4 mm) away from item C05 to minimize the effects of stray capacitance between the module and the bridge.
3. Do not attach test connector or wires to any of the module pins other than pins 10, 11, 30, 31, and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 24. Input capacitance test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

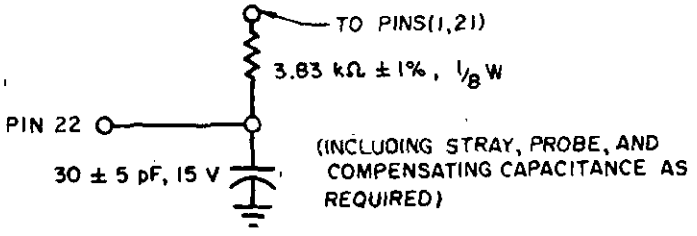


NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 21.5Ω ±1%, 1/4 W.
3. R8 = R9 = 50Ω ±5%, 1 W.
4. If a failure occurs for test condition S3, pulse width as measured per figure 26 may be as great as 1.3 µs.

FIGURE 25. Input transient immunity test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.

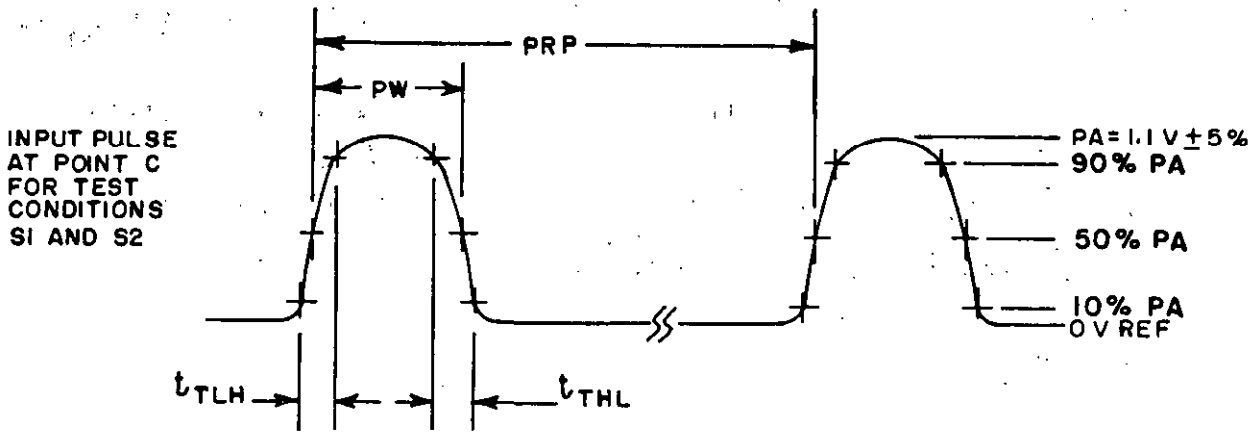
5. The impedance Z is composed of a 3.83 kΩ resistor from pin 22 to pin 1 and a 30 ±5 pF capacitance from pin 22 to ground as defined in note 7.



6. The loop length from pin 1 to pin 22, including the length of the body of the resistor, shall not exceed 4.5 inches (114.3 mm).
7. With the module removed from the test connector, measure the capacitance between the test connector pin 22 and pins 10, 11, 30,and 31 tied together. The measurements shall be performed at the point of mechanical contact between the module blade pin and the test connector pins. The measured value at 1 MHz ±10% plus probe capacitance shall be 30 ±5 pF. For these measurements, disconnect the 3.83 kΩ resistor from pin 22. The resistor should be disconnected as close to its body as practical. Reconnect the resistor prior to using the test connector for module testing.

FIGURE 25. Input transient immunity test circuit for table XI - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 20 ±2 ns.
PRP = 5.98 ±0.60 µs.
$t_{TLH} = t_{THL}$ = 6 ±1 ns.

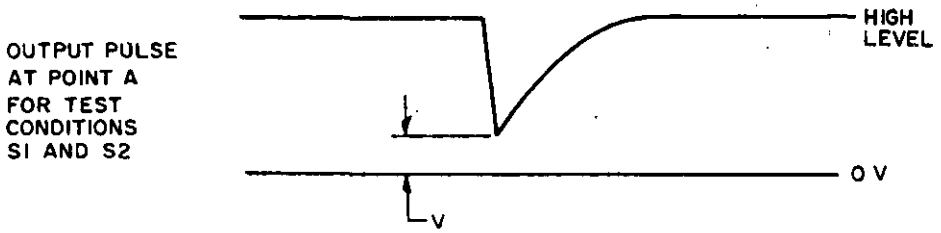
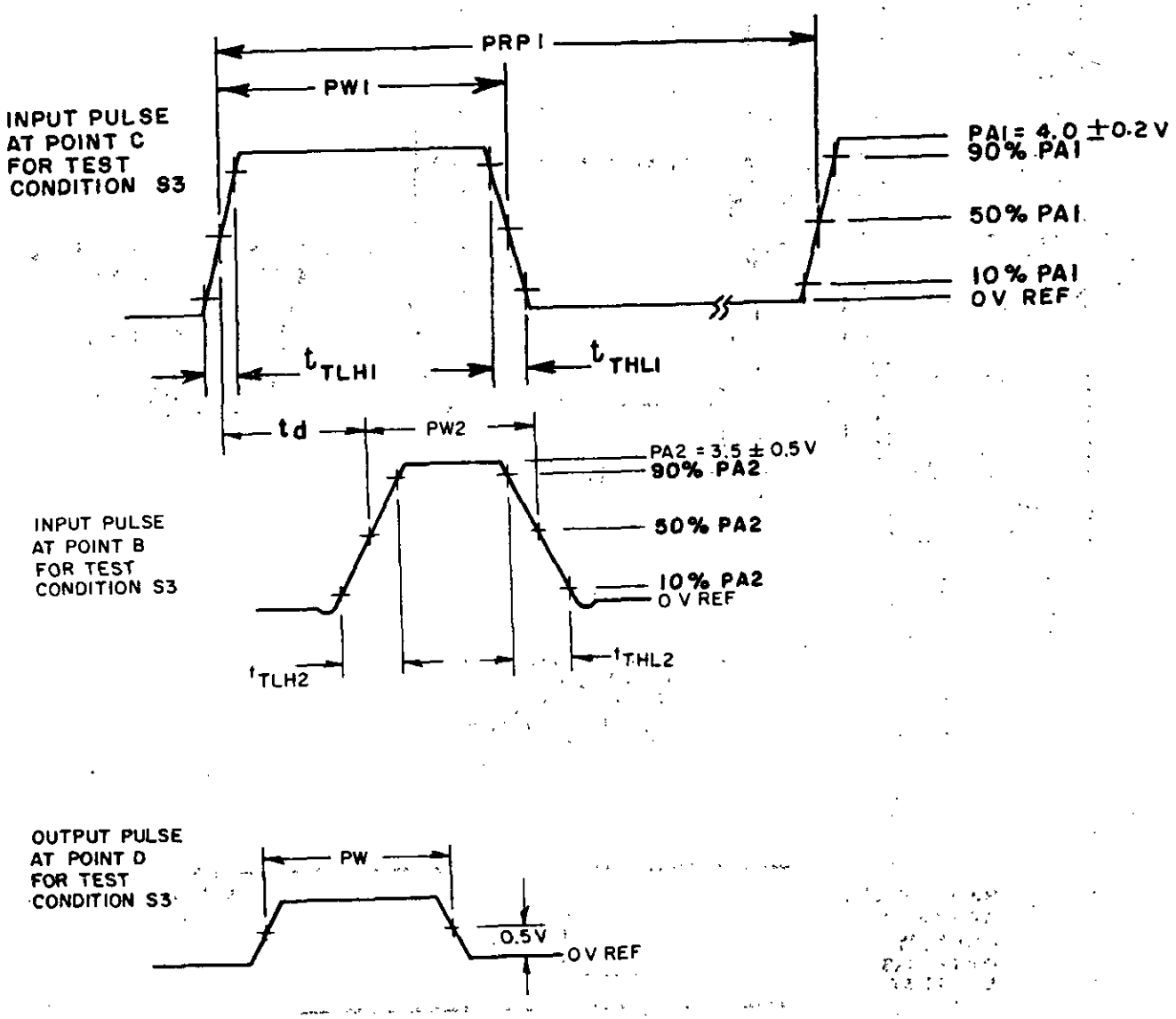


FIGURE 26. Input transient immunity test circuit waveforms for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW1 = 1.0 ±0.2 µs.
PRP1 = 5.98 ±0.60 µs.
$t_{TLH1} = t_{THL1}$ = 15 ±5 ns.
t_d = 0.5 ±0.1 µs.
PW2 = 30 ±5 ns.
$t_{TLH2} = t_{THL2}$ = 10 ±4 ns.

FIGURE 26. Input transient immunity test circuit waveforms for table XI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE XII. Input crosstalk test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Input crosstalk	3	24	0.0	0.6	V	27,28
T2	"	6	"	"	"	"	"
T3	"	9	"	"	"	"	"
T4	"	15	"	"	"	"	"
T5	"	17	"	"	"	"	"
T6	"	38	"	"	"	"	"
T7	"	20	"	"	"	"	"
T8	"	6	3	"	"	"	"
T9	"	9	"	"	"	"	"
T10	"	15	"	"	"	"	"
T11	"	17	"	"	"	"	"
T12	"	38	"	"	"	"	"
T13	"	20	"	"	"	"	"
T14	"	6,9,15,17,20, 38	22	2	5	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

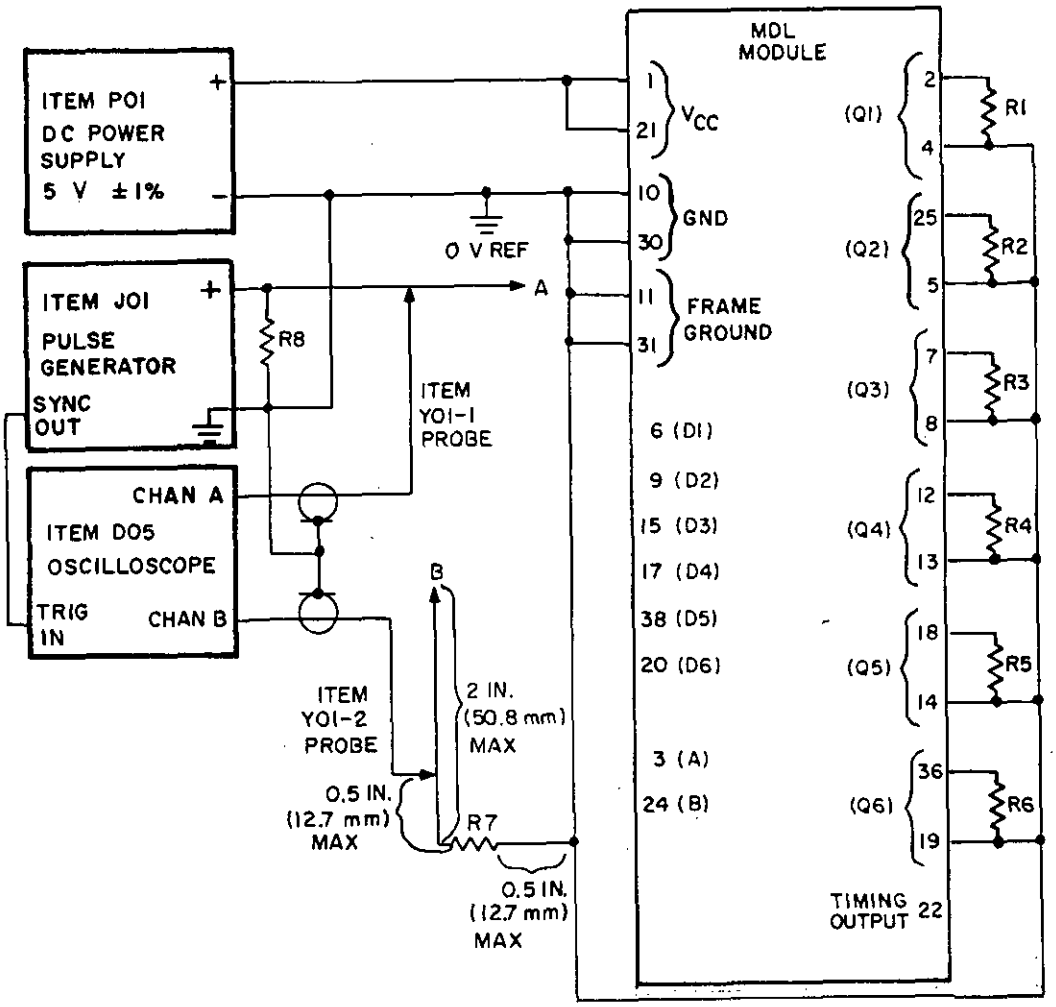


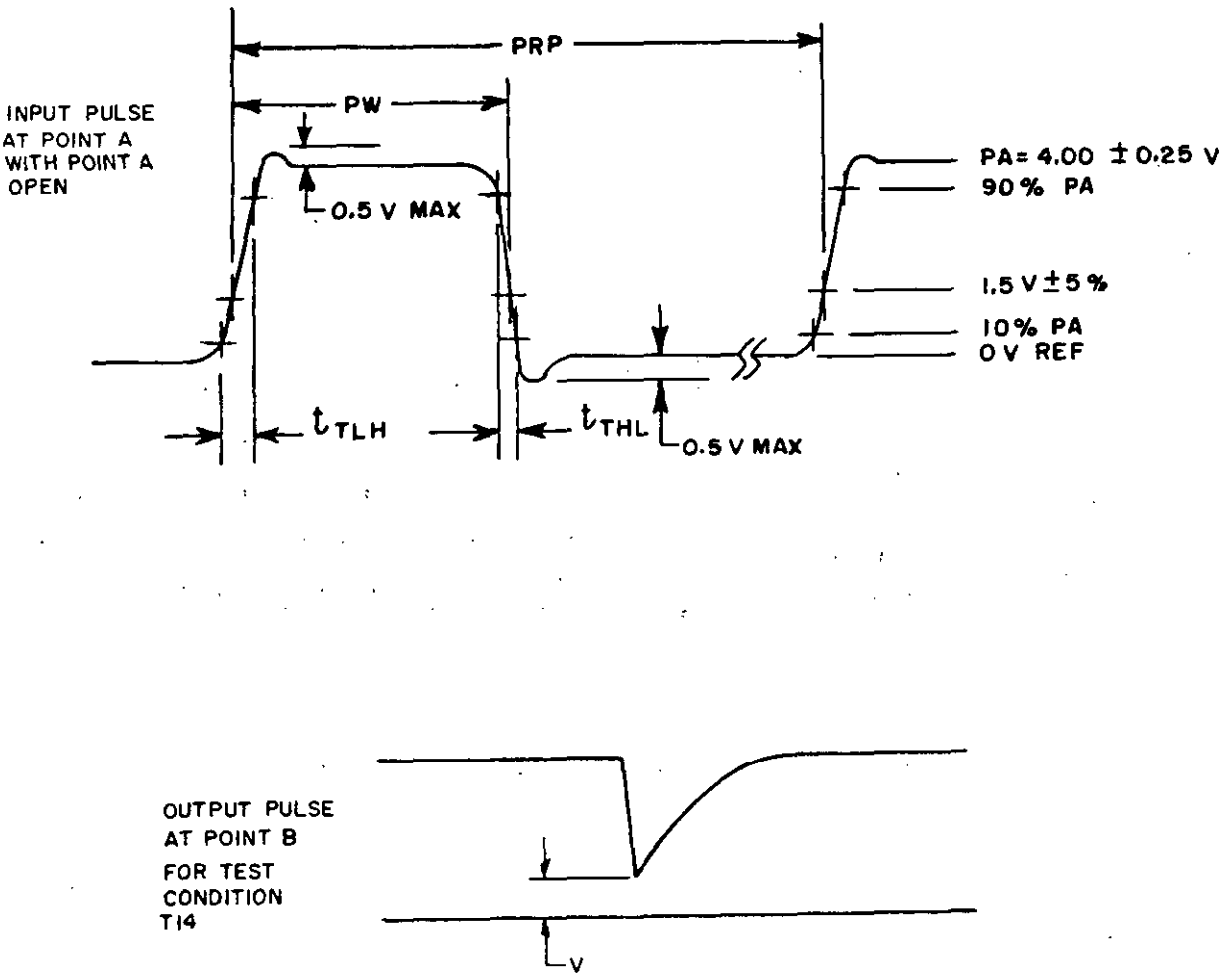
FIGURE 27. Input crosstalk test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 154Ω ±1%, 1/2 W.
3. R8 = 50Ω ±5%, 1 W.
4. Disconnect resistor R7 for test condition T14.
5. Item Y01-2 and its associated ground lead must be within 0.5 inch (12.7 mm) from the body of resistor R7.
6. The peak-to-peak voltage developed across resistor R7 must be equal to or less than 0.6 V for test conditions T1-T13.
7. The length of the conductor from the body of resistor R7 to point B shall not exceed 2.0 inches (50.8 mm).
8. The length of the conductor from the body of resistor R7 to the junction of pins 10, 11, 31, and 30 shall not exceed 2.0 inches (50.8 mm).
9. The maximum capacitance between adjacent input connectors on the test fixture shall not exceed 10 pF when measured at 1 MHz ±10% with the module removed from the test fixture.

FIGURE 27. Input crosstalk test circuit for table XII - Continued.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 300 ±30 ns.
PRP = 5.98 ±0.60 µs.
$t_{TLH} = t_{THL}$ = 10 ±1 ns.

FIGURE 28. Input crosstalk test circuit waveforms for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

TABLE XIII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply filter capacitance	0.10	0.26	µF	29

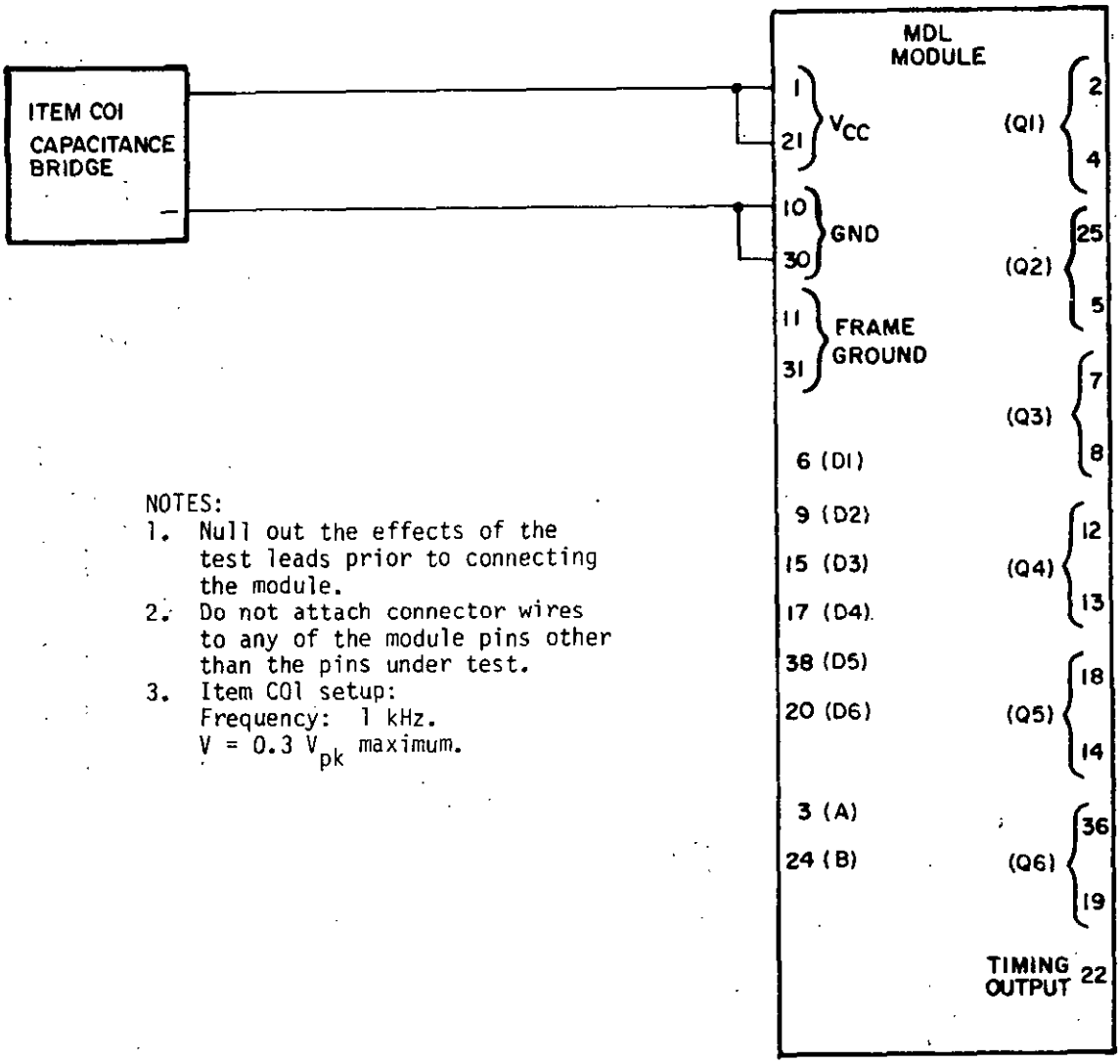
TABLE XIV. Output short and open test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Output short and open	0	150	mA	30,31

TABLE XV. Imbalance current test.

Test condition	Parameter	Connect		Item Y03 measuring current in connection between pins	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin		Min	Max.		
W1	Imbalance current	9,15,17,20,38	6	4 and 5	0	40	mA	32,33
W2	Imbalance current	6,9,17,20,38	15	8 and 13	0	40	mA	32,33
W3	Imbalance current	6,9,15,17,20	38	14 and 19	0	40	mA	32,33

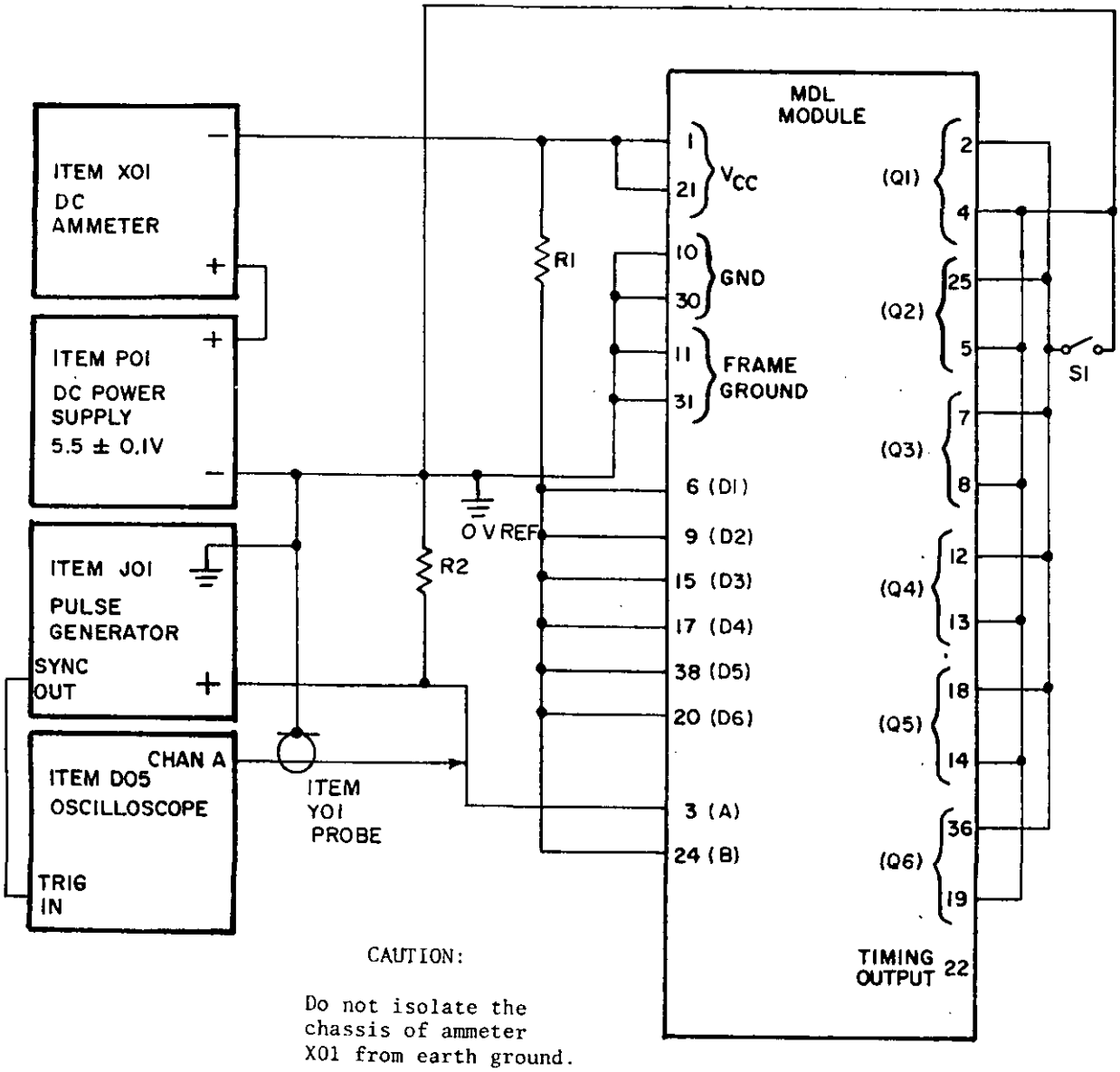
Source: https://assist.dla.mil -- Downloaded: 202640-20T16:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach connector wires to any of the module pins other than the pins under test.
3. Item COl setup:
   Frequency: 1 kHz.
   V = 0.3 V_{pk} maximum.

FIGURE 29. Power supply filter capacitance test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

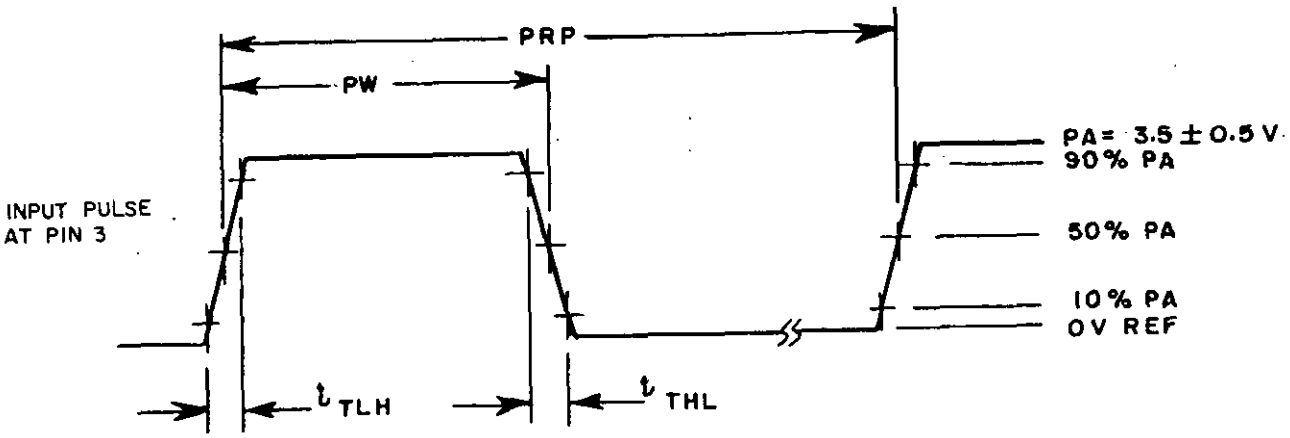


NOTES:
1. R1 = 3 kΩ ±1%, 1/8 W.
2. R2 = 50 ±5%, 1 W.
3. S1 = SPST switch.
4. Test sequence:

   a. Close switch S1 for 30 seconds minimum.
   b. Open switch S1.
   c. Measure current on item X01.

FIGURE 30. Output short and open test circuit for table XIV.

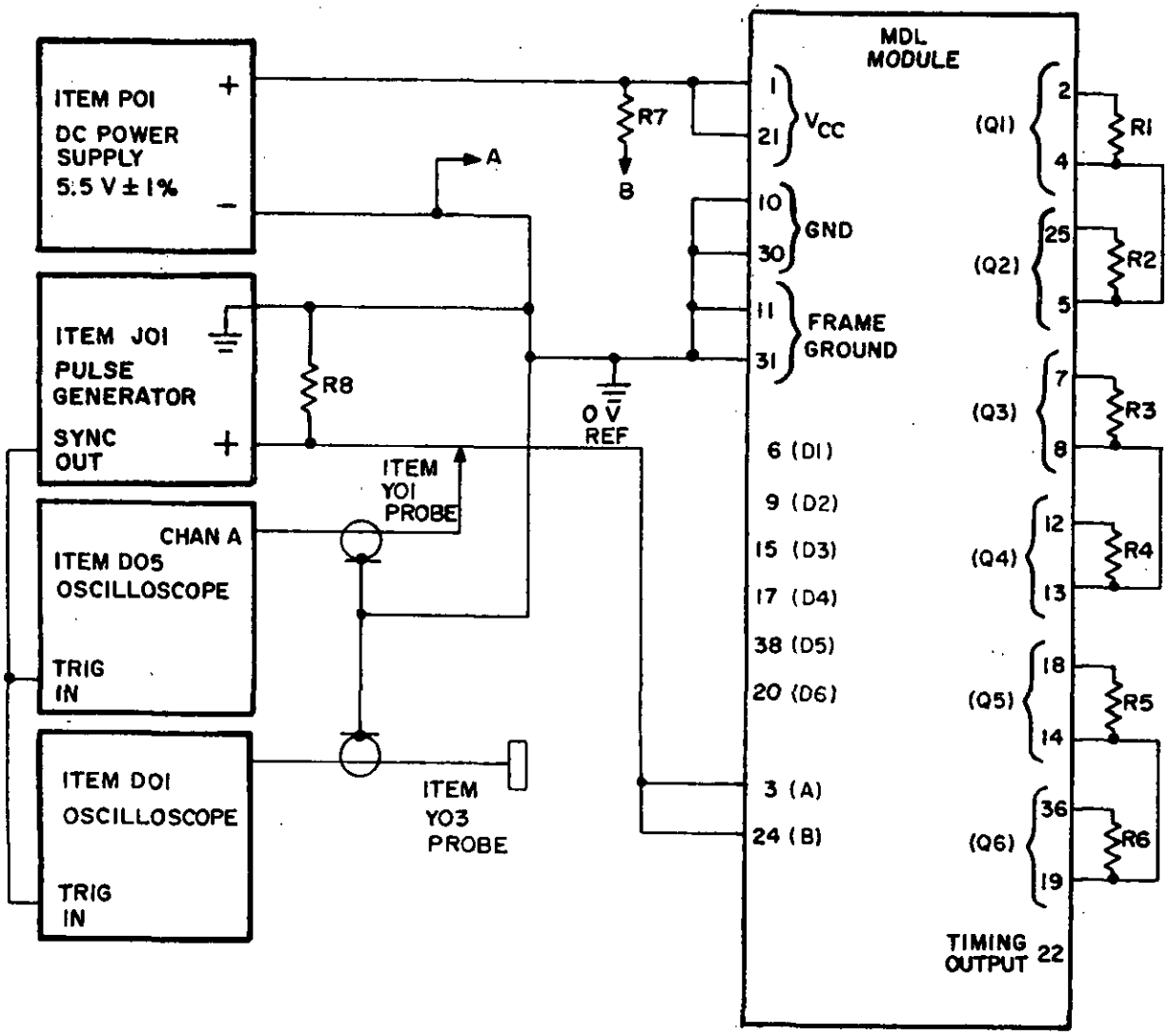
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 3.0 ±0.5 µs.
PRP = 15 ±3 µs.
$t_{TLH} = t_{THL}$ = 50 ±20 ns.

FIGURE 31. Output short and open test circuit waveform for table XIV.

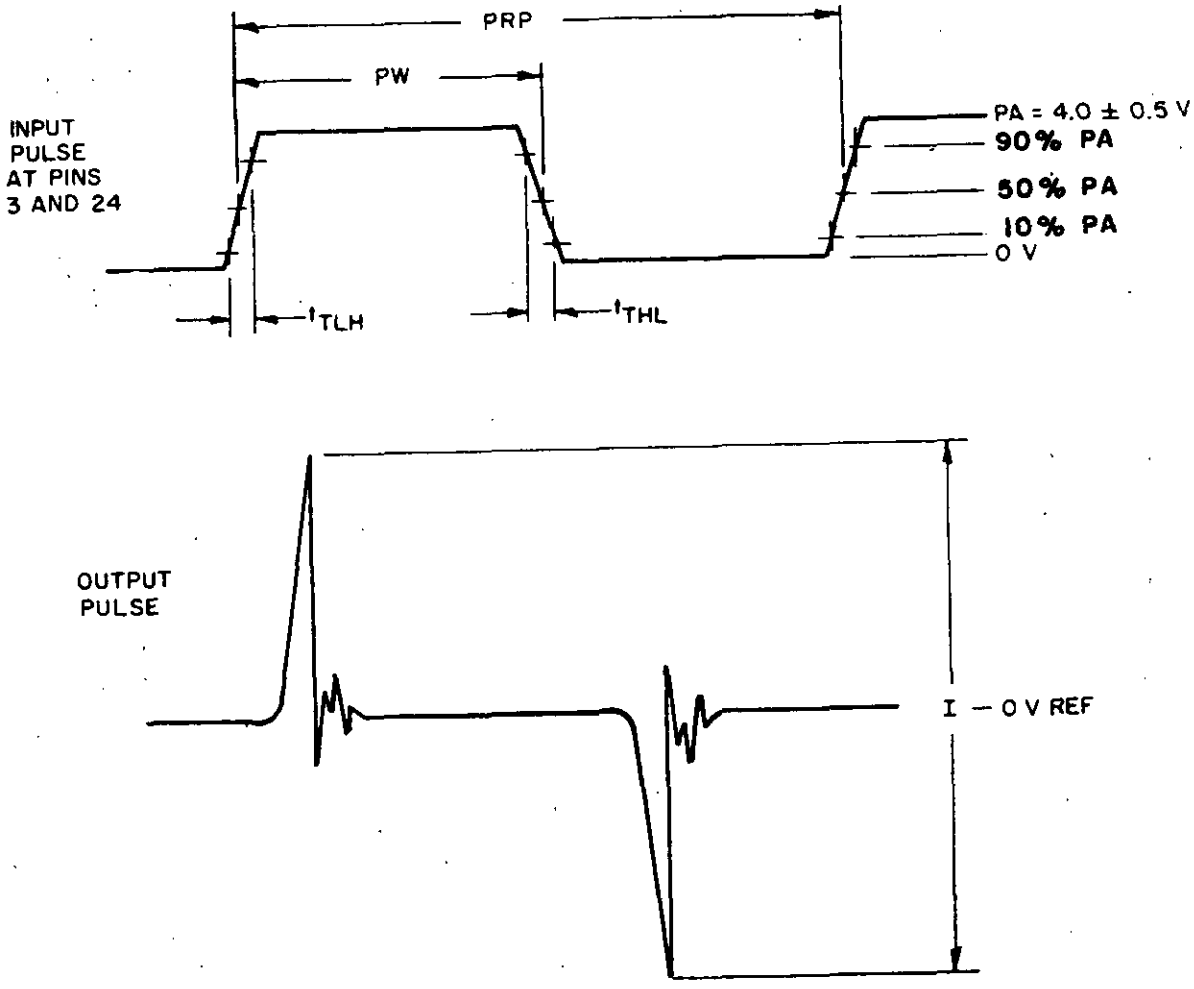
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R6 = 75Ω ±1%, 1/4 W.
2. R7 = 3 kΩ ±1%, 1/8 W.
3. R8 = 50Ω ±5%, 1 W.

FIGURE 32. Imbalance current test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PW = 3.0 ±0.5 µs.
PRP = 11 ±1 µs.
$t_{TLH} = t_{THL}$ = 20 ±10 ns.

FIGURE 33. Imbalance current test circuit waveforms for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/109B(EC) MODULES, STANDARD ELECTRONIC DRIVER, INTERFACE, DIGITAL, KEY CODE MDL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:03Z
Check the source to verify that this is the current version before use.